

Connected Intervention Calculus for Converter-Rich Power Systems: Re-Equilibrated Dynamic Externalities, Cycle Mechanisms, and Certified EMT Boundaries

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Abstract—Converter-rich power systems expose a planning problem that is not captured by asking whether each project, controller, or line is acceptable in isolation: finite engineering interventions can become harmful only after the nonlinear operating point is re-solved and the resulting network–controller feedback closes across several actions. This paper develops a connected intervention calculus for that problem. The mathematical foundation is rebuilt on the equilibrium manifold rather than on a frozen low-rank perturbation. If $H(y, a) = 0$ defines the AC operating point and $T(s, a)$ is the corresponding re-linearized dynamic pencil, the total action derivative contains both direct and equilibrium-mediated channels. An exact Jacobi transport identity then evaluates finite spectral change along any regular action path without assuming that the final perturbation is low rank.

The logarithm of the determinant is used as a connected generating functional. Mixed action derivatives obey a partition-cycle formula: higher-order re-equilibration terms appear as multi-action vertices, resolvents transport them through the network, and traces close the result into connected hypercycles. In the affine fixed-equilibrium limit, the formula collapses exactly to oriented controller-mediated action cycles. A second exact result shows that every finite Harsanyi/Möbius externality is the integral of the corresponding connected mixed derivative over the physical action cell. This unifies local differential interaction and finite portfolio non-additivity while keeping engineering allocation real-valued and branch free.

For low-rank port actions, the general theory reduces to a MIMO action quiver. Determinants are expanded exactly through finite cycle systems; logarithms are organized by closed walks and connected overlapping-cycle clusters, with recent quiver trace identities providing the proper prime-cyclic-walk language. Holonomy spectra are invariant to local proprietary state coordinates, and the algebra generated by fundamental holonomies determines whether a common scalar-channel representation exists. Certified sparsification is tied to decay of sparse resolvents rather than arbitrary edge thresholding.

The same operator structure produces a physically meaningful model-fidelity problem. A dangerous action core is mapped to a physical EMT region by preserving the exterior Schur self-energy seen at its boundary. The proposed minimal detailed region is the smallest region whose reduced exterior self-energy remains within a prescribed frequency-domain/structured-robustness tolerance. Complex-frequency strength, QSS frequency, and sinusoidality are retained as local temporal/waveform diagnostics; structured EMT-model uncertainty supplies an independent trust gate. Wavemaker

and endogeneity concepts distinguish visibility, receptivity, modal production, and closed feedback.

Mechanism-level IEEE 39-bus studies retain a strong planning counterexample: closing a three-GFL collector loop improves retained-port Fiedler connectivity by 43.9% while minimum damping falls from 5.152% to 2.876%; a radial dual-POI redesign restores 5.326%. The contribution is therefore not another universal grid-strength index. It is a finite-action theory for determining when individually acceptable interventions fail to compose, identifying the connected controller–network mechanism, preserving that mechanism in the minimum detailed EMT model, and computing the smallest physically realizable recovery action.

Index Terms—inverter-based resources, dynamic interactions, finite interventions, equilibrium manifold, connected cumulants, action quiver, holonomy, electromagnetic transient studies, model reduction, Schur complement, self-energy, complex frequency, grid strength, interconnection planning, Shapley value.

I. INTRODUCTION

CONVERTER-RICH grids have made a familiar engineering workflow increasingly fragile. A candidate line is screened, a controller is tuned, a BESS is added, and each modification is judged individually. Yet the implemented system is not the sum of those isolated studies: every finite change can move the nonlinear AC equilibrium, modify converter operating points, reshape network transfer paths, and close feedback through other devices. The relevant question is therefore not merely “which bus is weak?” but

When do finite engineering actions fail to compose, which connected network–controller mechanism creates the non-additive effect, and what is the smallest model/control change that removes it?

The question sits between several mature tool families. Eigenvalue and participation analysis identify modes and states; impedance/return-ratio and multiport dissipativity methods diagnose converter coupling; graph and grid-strength metrics characterize local/network support; complex frequency provides a physically meaningful magnitude–angle dynamical coordinate; stability-manifold methods map admissible controller regions; EMT simulation resolves dynamics omitted by phasor models. None of these should be portrayed as deficient. The narrower gap pursued here is *finite portfolio composition*: the joint effect of named engineering actions after the actual operating point is re-solved, and the mapping of that mechanism to the smallest detailed model that must preserve it.

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That distinction matters industrially. Recent work on converter-dominated stability shows that interactions among multiple IBRs can reshape admissible controller regions and that simplified equivalents can miss system-level limits [69]. Recent phasor-model-validity work formulates omitted EMT dynamics as structured uncertainty and uses robust-stability certificates to decide when a reduced device model can be trusted [19]. EMT-transient-stability hybrid work identifies interface error and region expansion as central accuracy issues [58]. At the event level, a 2024 Texas large-electronic-load incident produced approximately 23-Hz oscillations near 320 MW and was reproduced in detailed EMT studies [59]. The planning need is consequently not another scalar replacement for EMT, but a mechanistic triage layer that can say which interaction must survive the detailed model.

A. Why the mathematical foundation is rebuilt

An earlier version of this program used a low-rank finite-action update

$$T_A = T_0 + UCV^*$$

as the master representation. That identity is valuable when the action is applied to a frozen linear operator, but it is not a general model of a real re-equilibrated line or controller change. Re-solving power flow changes the equilibrium and may change the linearization globally. Rather than patch that limitation, this paper moves the foundation one level deeper:

$$\boxed{\begin{aligned} H(y, a) = 0 &\longrightarrow y^* = \chi(a) \\ \longrightarrow T(s, a) &\longrightarrow \text{connected mixed action calculus} \end{aligned}} \quad (1)$$

Low-rank/quiver structure is then recovered as an efficient tangent special case.

This change also resolves several other conceptual issues at once. It separates simple cycle covers from primitive closed walks; replaces a complex-valued cooperative game by real finite dividends; distinguishes one-cycle closure from overlapping-cycle clusters; replaces a coordinate-dependent “non-Abelian score” by the generated holonomy algebra; gives action-graph sparsification a locality certificate; and maps an action core to a physical EMT boundary through exterior self-energy rather than by equating an SCC with a set of buses.

B. Main contributions

The paper makes five tightly connected contributions.

- 1) *Re-equilibrated finite-action calculus.* The operating point is treated as an implicit equilibrium manifold. Exact total action derivatives and a finite spectral-transport identity remain valid when a physical intervention changes the entire re-linearized operator.
- 2) *Connected interaction theorem.* Mixed derivatives of $\log \det T$ are written as a partition-cycle expansion. In the affine limit they become exact closed action-cycle traces; finite Harsanyi/Möbius externalities are exact integrals of these connected densities.
- 3) *Cycle/cluster and MIMO structure.* The determinant is separated into finite cycle-cover physics and connected closed-walk/cluster physics. MIMO action holonomies are

organized by a coordinate-invariant holonomy algebra that determines common scalarizability.

- 4) *Certified localization.* Sparse-resolvent decay motivates frequency-dependent action sparsification with perturbation certificates, addressing the otherwise serious possibility of one giant dense SCC.
- 5) *Cycle-preserving EMT boundary.* A dangerous interaction core is mapped to a physical detailed region by preserving the Schur self-energy of the omitted exterior, combined with waveform and structured model-mismatch trust gates. The mathematical identities are not presented as new linear algebra in isolation. The candidate power-systems novelty is their composition into a finite-intervention framework and the specific claim that connected dynamic externalities can be localized, attributed, and preserved across RMS-to-EMT model boundaries.

II. LITERATURE BOUNDARY AND CLOSEST COMPETING TOOL FAMILIES

The nearest literature must be confronted directly. Multiport impedance, return-ratio, and dissipativity approaches already diagnose coupling and can suggest mitigation; recent SST work explicitly performs multiport root-cause identification and controller redesign [67]. Complex-frequency strength quantifies zero-, first-, and second-order resistance of bus-voltage dynamics to injection changes [46]. Stability-manifold methods map controller/operating-point safe regions under multi-IBR interaction [69]. Graph-theoretic and perturbation-space methods analyze topology-dependent stability and interaction sensitivity [16]–[18]. Therefore the contribution cannot be “we identify converter interactions.”

The proposed gap is more specific:

$$\boxed{\begin{aligned} &\text{diagnosis of one configuration} \\ &\neq \text{composition law for finite re-equilibrated} \\ &\quad \text{action portfolios} \end{aligned}} \quad (2)$$

The paper asks whether individually acceptable actions remain additive/composable, decomposes the irreducible finite effect by connected action sets, and then uses the identified mechanism to define a physical model boundary. Table I states the novelty boundary.

III. EXACT RELATIONSHIP TO RETURN-RATIO AND IMPEDANCE STABILITY METHODS

A reviewer familiar with multivariable control should recognize that the low-rank action operator is closely related to a generalized return ratio. Making that relationship explicit sharpens, rather than weakens, the novelty claim.

Consider the affine port update

$$T_A = T_0 + UCV^*, \quad G_0 = T_0^{-1}. \quad (3)$$

The determinant lemma gives

$$\frac{\det T_A}{\det T_0} = \det(I + K), \quad K = CV^*G_0U. \quad (4)$$

This is exactly the algebraic structure underlying multiloop return-ratio and generalized-Nyquist analyses: the full-system

TABLE I
CLAIM BOUNDARY RELATIVE TO NEIGHBORING POWER-SYSTEM AND MATHEMATICAL TOOLS

Tool family	Established capability	What is used here	What is not claimed
Impedance / return ratio / dissipativity	Converter coupling, frequency-domain stability, multiport root cause, mitigation.	Port-level transport and robust baselines.	Interaction localization itself is not claimed as new.
Complex frequency / strength	Local voltage magnitude-angle dynamics, synchronization/coherency, dynamic strength orders.	Observable local temporal layer and benchmark.	CF/strength is not relabeled as our stability metric.
Stability manifolds / sensitivities	Safe controller regions and local parameter/topology derivatives.	Action-space boundary geometry and local normals.	Mapping a stability boundary is not our novelty.
Schur / self-energy / model reduction	Exact elimination and external equivalents.	Exterior self-energy as the quantity a reduced EMT boundary must preserve.	Schur complements are not new.
Determinant/cumulant/cluster mathematics	Log determinants generate connected traces; determinants admit cycle-cover interpretations.	Pullback to the AC equilibrium manifold and intervention/hypercycle interpretation.	The generic combinatorial identities are not claimed as new mathematics.
This work	—	Finite re-equilibrated non-additivity, connected action cumulants, low-rank cycle realization, and mechanism-preserving EMT boundary.	No universal scalar strength/stability index.

pole change is encoded by a low-dimensional feedback matrix whose determinant/winding carries the closed-loop information. Impedance/admittance formulations produce analogous loop matrices from terminal port relations, and recent impedance-modal methods can localize converter/control participation without requiring internal state access [63], [64].

Proposition 1 (Action-quiver/return-ratio equivalence). *For a fixed affine operating-point model, the action-quiver determinant $\det(I + K)$ is a generalized return-ratio determinant. Its argument-principle winding relative to the baseline equals the net change in zero/eigenvalue count, subject to the standard pole bookkeeping. Local port-coordinate changes induce the block similarities described in (68) and leave the determinant and holonomy characteristic invariants unchanged.*

Proof. Equation (4) is the matrix determinant lemma. Taking the winding of both sides on a contour gives the difference of logarithmic-residue counts. Port-basis invariance follows by conjugating the corresponding return blocks. $\square$

The paper therefore does *not* claim a new Nyquist criterion. The distinction is the object to which the return-ratio algebra is applied. Conventional impedance/return-ratio analysis asks whether one assembled configuration is stable and which devices/control loops participate in the problematic loop. The connected intervention calculus introduces physical *action coordinates*, pulls the entire dynamic operator back to the re-equilibrated AC operating manifold, and asks whether finite named actions compose. The mixed derivatives and finite Möbius dividends then quantify irreducible project-set effects that are not obtained by analyzing only the final configuration.

This distinction is especially clear for three actions. A conventional final-configuration loop matrix can diagnose an unstable mode in the joint system. The connected calculus additionally tests whether every singleton and pair was acceptable, computes the genuine triple dividend, and separates that dividend into re-equilibrium vertices and pure feedback returns by Theorem 7. The final loop diagnosis and the portfolio-composition diagnosis are complementary, not competing, analyses.

A. What the new cycle language adds and what it does not

Impedance methods already expose closed converter-grid feedback and can suggest retuning. Extended impedance modal analysis can separate PLL- and current-loop contributions [64], while recent Pauli decompositions seek a more interpretable physical basis for MIMO impedance coupling [65]. Accordingly, phrases such as “existing methods cannot identify controller interactions” are excluded from this manuscript.

The new cycle language is used for three narrower purposes: (i) to label *intervention* cycles rather than only device-port loops, (ii) to organize higher-order finite non-additivity through connected cycle clusters and dividends, and (iii) to link the identified action core to an exterior-self-energy model boundary. If an impedance/dissipativity method provides the same final-config root cause more directly, it should be used as a baseline and its diagnosis should agree with the relevant connected core.

IV. FROM THE AC-DAE TO A CONTROLLER-AWARE DYNAMIC OPERATOR

A. Physical starting point

Let the nonlinear network and devices be represented by

$$\dot{x} = f(x, z, p), \quad 0 = g(x, z, p), \quad (5)$$

where x are dynamic states, z algebraic variables, and p parameters or actions. At a regular operating point, eliminating algebraics yields the standard reduced linearization

$$A_{\text{red}} = f_x - f_z g_z^{-1} g_x. \quad (6)$$

Under restrictive lossless, stiff-voltage, small-angle assumptions, the electromechanical part may be summarized by

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \quad (7)$$

with inertia M , damping D , and operating-point-dependent active-power Jacobian L . The project initially explored whether L , or its mass-normalized spectrum, was enough to define a dynamic hosting margin. It is not: L measures restoring stiffness, while damping and controller states determine decay and interaction.

B. Exact controller elimination

Partition a full linearization into retained network/interface states x_r and internal controller/auxiliary states x_c :

$$\begin{bmatrix} \dot{x}_r \\ \dot{x}_c \end{bmatrix} = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix} \begin{bmatrix} x_r \\ x_c \end{bmatrix} + \begin{bmatrix} B_r \\ B_c \end{bmatrix} u. \quad (8)$$

For $sI - A_{cc}$ invertible,

$$x_c = (sI - A_{cc})^{-1}(A_{cr}x_r + B_c u), \quad (9)$$

and the retained dynamics satisfy

$$S_r(s)x_r = B_{\text{eff}}(s)u, \quad (10)$$

$$S_r(s) = sI - A_{rr} - \Sigma_A(s), \quad (11)$$

$$\Sigma_A(s) = A_{rc}(sI - A_{cc})^{-1}A_{cr}. \quad (12)$$

The block determinant identity

$$\det(sI - A) = \det(sI - A_{cc}) \det S_r(s) \quad (13)$$

shows that, with standard bookkeeping at internal poles, the Schur nonlinear eigenvalue problem reproduces the full modes that participate in the retained coordinates. The word *self-energy* is useful because the eliminated controller states leave a frequency-dependent feedback footprint on the retained network, analogous to a memory kernel.

When the retained coordinates admit a second-order representation, the project uses the master dynamic stiffness

$$T(s; \rho) = s^2 M(\rho) + sD(\rho) + L(z^*(\rho), \rho) + \Pi(s, z^*(\rho), \rho), \quad (14)$$

where Π is the controller-induced correction after consistent elimination. An equivalent notation absorbs D into a generalized matrix $\Sigma(s)$ and writes $T = s^2 M + s\Sigma(s) + L$. Both conventions encode the same idea: controller dynamics need not be local, diagonal, symmetric, or frequency independent after the network is closed.

C. L - M - D as a low-frequency approximation

If the eliminated controller block is stable and $|s|$ is small relative to its internal scales,

$$(sI - A_{cc})^{-1} = -A_{cc}^{-1} - sA_{cc}^{-2} - s^2 A_{cc}^{-3} - \dots, \quad (15)$$

so locally

$$\Pi(s) \approx \Pi_0 + s\Pi_1 + s^2\Pi_2 + \dots \quad (16)$$

The first coefficients can be absorbed into effective stiffness, damping, and inertia. This is an important conceptual correction: an L - M - D model is not “wrong” by definition; it is a truncated representation whose trust region must be diagnosed. The framework therefore seeks *gates*, not a mandatory full-order computation at every operating point.

D. Controller-induced nonlocality

Even if a controller is implemented locally, $\Pi_{ij}(s)$ can be nonzero for distinct retained buses after algebraic closure and state elimination. Local hardware can therefore generate an effective network interaction. This is the first reason that a fixed nodal damping coefficient can point to the wrong mechanism: the quantity replacing it is a matrix transfer function.

V. SUPPORTING DIAGNOSTICS: STIFFNESS, DAMPING, HYBRIDIZATION, AND DIRECTED RESPONSE

The connected calculus does not discard the earlier diagnostics; it assigns them a narrower role. A mass-normalized Laplacian describes restoring stiffness, not universal dynamic security. A damping-region/QEP gate diagnoses when modal damping cannot be treated diagonally. A physical resolvent identifies directed forcing/response amplification. Biorthogonal mode-family participation is used to detect network-controller hybridization without arbitrary state-unit scaling. These objects are retained as explanatory coordinates and baselines, not as headline stability margins.

For $Lq_k = \nu_k M q_k$ with $q_k^T M q_k = 1$, inertia sensitivity obeys $\partial \nu_k / \partial M_i = -\nu_k q_{k,i}^2$, so lower inertia can raise natural frequency without improving damping. For a simple nonlinear eigenvalue $T(s, \rho)v = 0$, $w^* T = 0$,

$$\frac{ds}{d\rho} = -\frac{w^* T_\rho v}{w^* T_s v}. \quad (17)$$

Near network-controller modal hybridization the denominator can become ill-conditioned, amplifying topology/controller sensitivity. This is the local differential precursor to the finite connected effects developed in Sections VII–VIII.

VI. NETWORK-CONTROLLER HYBRIDIZATION AND RESONANT TOPOLOGY SENSITIVITY

A. Scale-invariant mode-family participation

Raw state norms are coordinate dependent: rescaling a PLL integrator changes its norm contribution without changing physics. For right and left eigenvectors r, l , define the controller-family fraction

$$\pi_c^{\text{bio}} = \frac{\sum_{i \in \mathcal{C}} |\bar{l}_i r_i|}{\sum_i |\bar{l}_i r_i|}. \quad (18)$$

Under an invertible diagonal state transformation $x' = Sx$, $r' = Sr$ and $l' = S^{-H}l$, so each product $\bar{l}'_i r'_i = \bar{l}_i r_i$. Hence (18) is invariant to diagonal state-unit scaling. In the numerical coordinate-scaling test, a norm-based controller fraction moved from nearly zero to nearly one as controller coordinates were scaled from 10^{-6} to 10^6 , while the biorthogonal fraction stayed at 0.597033.

B. Two-mode hybridization model

Let an uncoupled network mode λ_n interact with a controller mode λ_c through gains g_{nc}, g_{cn} :

$$H_{nc} = \begin{bmatrix} \lambda_n & g_{nc} \\ g_{cn} & \lambda_c \end{bmatrix}, \quad \lambda_{\pm} = \frac{\lambda_n + \lambda_c}{2} \pm \sqrt{\frac{(\lambda_n - \lambda_c)^2}{4} + g_{nc}g_{cn}}. \quad (19)$$

A dimensionless diagnostic is

$$\mathcal{H}_{nc} = \frac{4|g_{nc}g_{cn}|}{|\lambda_n - \lambda_c|^2}. \quad (20)$$

It is not proposed as a universal stability criterion. Its role is to signal that a network/control family separation is becoming unreliable.

Differentiating (19) shows why action sensitivity can become large as the square-root discriminant approaches zero. In a mechanism sweep, a near-coalescent pair had separation 0.02125, controller-family fractions 0.478/0.521, and eigenvalue condition numbers near 73.5. The maximum line-action sensitivity was 2.05×10^{-2} , compared with 4.10×10^{-4} in a detuned case—about $50\times$ larger. Across 527 sampled points, the Spearman association between eigenvalue conditioning and maximum absolute topology sensitivity was 0.593.

C. Exact nonlinear-eigenvalue derivative

For a simple nonlinear eigenvalue $s(\rho)$ satisfying

$$T(s, \rho)v = 0, \quad y^*T(s, \rho) = 0, \quad (21)$$

standard differentiation gives

$$\frac{ds}{d\rho} = -\frac{y^*T_\rho v}{y^*T_s v}. \quad (22)$$

For $T = s^2M + s\Sigma(s) + L$,

$$\frac{ds}{d\rho} = -\frac{y^*(s^2M_\rho + s\Sigma_\rho + L_\rho)v}{y^*(2sM + \Sigma + s\Sigma_s)v}. \quad (23)$$

The denominator is the controller-aware renormalization omitted by a constant-damping QEP. If $s = \sigma + j\omega$ and $s_\rho = a + jb$,

$$\frac{d\zeta}{d\rho} = \frac{\sigma\omega b - \omega^2 a}{|s|^3}. \quad (24)$$

D. Structural, power-flow, and controller channels

If the operating point x^* satisfies $F(x^*, z) = 0$ and F_x is nonsingular,

$$\frac{dx^*}{dz_e} = -F_x^{-1}F_{z_e}. \quad (25)$$

The total topology derivative can be written schematically as

$$\begin{aligned} \frac{dT}{dz_e} = & \underbrace{L_{z_e}}_{\text{structural}} - \underbrace{L_x F_x^{-1} F_{z_e}}_{\text{flow-mediated}} \\ & + s \underbrace{(\Sigma_{z_e} - \Sigma_x F_x^{-1} F_{z_e})}_{\text{controller-mediated}}. \end{aligned} \quad (26)$$

In one mechanism test, line 5–8 had a structural-only derivative of $+3.33 \times 10^{-7}$ but a derivative of -1.40×10^{-5} after AC power flow was re-solved. A stronger controller-resonant case occurred for line 25–26: the network-only damping sensitivity was $+1.3408 \times 10^{-4}$, while the controller-aware sensitivity was -4.2715×10^{-3} . The sign reversed and the magnitude increased by about $31.9\times$. Analytic and finite-difference derivatives agreed to relative error on the order of 10^{-8} for well-conditioned steps.

These results motivate the term *controller-resonant topology action*: a topology modification whose electrical stiffness interpretation is altered because it moves a network mode toward or away from a controller family. The novelty is not the existence of modal resonance [12], [13]; it is the planning chain from an exact self-energy to a changed finite-action decision.

VII. CONNECTED INTERVENTION CALCULUS ON THE RE-EQUILIBRATED OPERATING MANIFOLD

The central revision of this manuscript is conceptual rather than cosmetic. A physical power-system intervention is not, in general, a low-rank perturbation of a frozen linear model. A line addition, point-of-interconnection change, controller retuning, or converter replacement changes the nonlinear equilibrium and therefore changes the entire linearized descriptor operator. The correct exact object is consequently the *pullback of the dynamic pencil to the equilibrium manifold*. Low-rank action matrices and action quivers then appear as a useful special case, not as the foundation.

A. Equilibrium manifold and pullback dynamic pencil

Let $a \in \mathcal{A} \subset \mathbb{R}^m$ denote physically interpretable actions and let y collect all steady-state differential and algebraic variables. The operating point satisfies

$$H(y, a) = 0. \quad (27)$$

Assume H is C^k and H_y is nonsingular in a neighborhood of (y_0, a_0) . The implicit-function theorem gives a unique local equilibrium map

$$y^* = \chi(a), \quad H(\chi(a), a) = 0. \quad (28)$$

Let $\mathcal{T}(s, y, a)$ denote the regularized descriptor/dynamic-stiffness operator obtained from the full AC-DAE after any exact algebraic elimination that does not change its finite eigenvalues. The action-dependent pencil is the pullback

$$T(s, a) = \mathcal{T}(s, \chi(a), a). \quad (29)$$

Every topology/control intervention used later in the IEEE 39-bus experiments is interpreted through (29): the AC power flow is re-solved first and the dynamic operator is then re-linearized at the new equilibrium.

Proposition 2 (Total action derivative). *Under the regularity assumptions above,*

$$D_i T = \mathcal{T}_{a_i} - \mathcal{T}_y H_y^{-1} H_{a_i}. \quad (30)$$

The first term is the direct structural/controller action; the second is the equilibrium-mediated correction.

Proof. Differentiate $H(\chi(a), a) = 0$ with respect to a_i to obtain $\chi_{a_i} = -H_y^{-1} H_{a_i}$. Applying the chain rule to (29) yields (30). $\square$

Equation (30) replaces the informal distinction between “direct” and “power-flow” effects by an exact differential on the physical operating manifold. Higher derivatives $D_B T$ for an action subset B follow by repeated implicit differentiation. Those higher derivatives are generally nonzero even when each device modification is structurally local: nonlinear re-equilibration itself can create multi-action coupling.

TABLE II
REPRESENTATIVE ACTION COORDINATES AND THEIR PULLBACK CHANNELS

Action	Direct channel	Re-equilibrium channel
Line/transformer parameter	Local admittance/topology block	Global V, θ , flows, converter operating points
PLL/current-loop gain	Local controller Jacobian	Often small at first order; mixed with loading/topology
P/Q setpoint or large electronic load	Device/load injection	Strong AC operating-point redistribution
Virtual inertia/droop	Local dynamic block	Depends on control equilibrium and power sharing
POI/collector design	Sparse branch homotopy	Global voltage/flow redistribution
GFL/GFM mode choice	Controller-family endpoint	Use smooth homotopy only when physically meaningful

B. Power-system realization of the action coordinates

The abstract coordinate a_i is chosen to correspond to an implementable engineering degree of freedom. The pullback derivative makes clear which actions are genuinely local and which become global only after re-equilibration.

For a transmission branch $e = (p, q)$ with series admittance parameter y_e , the bus-admittance contribution has the incidence form

$$Y_e = y_e b_e b_e^T, \quad (31)$$

where $b_e = e_p - e_q$. Hence the *direct* network derivative with respect to y_e is rank one (or a small real block after splitting real/imaginary coordinates). However, the solved angles and magnitudes satisfy

$$\chi_{y_e} = -H_y^{-1} H_{y_e}, \quad (32)$$

and therefore the full dynamic derivative is

$$D_{y_e} T = \mathcal{T}_{y_e} + \mathcal{T}_y \chi_{y_e}. \quad (33)$$

The first term is the familiar local topology update; the second can be dense because the entire AC operating point moves. This one equation is the precise reason the old “a line is a low-rank finite perturbation” foundation was insufficient.

A controller gain k_i often has the opposite structure: $\mathcal{T}_{k_i}$ is localized to one converter block, while H_{k_i} may vanish if the gain does not change the steady-state control equilibrium. In that special case the first-order action is genuinely local, but mixed derivatives with a line, loading level, or another controller can still be nonzero through $D_{ij} T$. A power-setpoint or large-load action directly changes H and therefore carries a strong equilibrium-mediated channel even when its dynamic device equations are unchanged.

Discrete design choices are embedded by a regular homotopy when a derivative-based mechanism is desired. A line opening/closing uses a continuous branch-strength parameter $\alpha_e \in [0, 1]$; a GFL-to-GFM comparison may use a physically meaningful controller interpolation only if intermediate models remain well posed. When no credible smooth interpolation exists, the framework does not invent one: endpoint security metrics and finite Möbius dividends remain valid, while derivative/hypercycle attribution is restricted to the smooth subfamilies.

This action dictionary makes the theory operational: every reported connected dividend must be traceable to named engineering coordinates, and every coordinate must state whether its derivative is analytic, implicitly differentiated, numerically identified, or endpoint-only.

C. Exact finite spectral transport without a low-rank assumption

The logarithmic derivative of a determinant gives an exact finite-action identity.

Theorem 3 (Re-equilibrated spectral transport). *Let $a : [0, 1] \rightarrow \mathcal{A}$ be a C^1 action path for which $T(s, a(\tau))$ is nonsingular at a fixed complex s . Then*

$$\log \frac{\det T(s, a(1))}{\det T(s, a(0))} = \int_0^1 \text{tr} \left[T^{-1}(s, a(\tau)) \dot{T}(s, a(\tau)) \right] d\tau, \quad (34)$$

for any continuous branch of the logarithm along the path. The branch-free magnitude identity is

$$\log \left| \frac{\det T(s, a(1))}{\det T(s, a(0))} \right| = \int_0^1 \Re \text{tr} \left[T^{-1} \dot{T} \right] d\tau. \quad (35)$$

Proof. Jacobi’s formula gives $\frac{d}{d\tau} \log \det T = \text{tr}(T^{-1} \dot{T})$. Integration proves (34); taking real parts gives (35). $\square$

Theorem 3 is exact for a re-solved AC equilibrium and does not require $T(a_1) - T(a_0)$ to be low rank. It is the global finite-action backbone of the paper. The familiar determinant lemma is recovered later when the pullback operator happens to be affine and low rank.

D. Stability boundary as an action-space discriminant

For a security contour Γ in the complex- s plane, define the action-space spectral discriminant

$$\mathfrak{D}_\Gamma = \{a : \exists s \in \Gamma, \det T(s, a) = 0\}. \quad (36)$$

The set $\mathfrak{D}_\Gamma$ is the physical boundary at which a pole reaches the chosen damping/stability contour. Recent stability-manifold work maps such boundaries numerically across controller/operating-point coordinates [69]. The present theory asks a different question: *which connected multi-action terms bend that boundary, and which feedback paths create those terms?*

If $s(a)$ is a simple nonlinear eigenvalue with right/left null vectors v, w , then

$$\frac{\partial s}{\partial a_i} = -\frac{w^* D_i T v}{w^* T_s v}, \quad (37)$$

so the normal to a smooth piece of $\mathfrak{D}_\Gamma$ is directly computable from the same pullback derivative (30). This connects local sensitivity, finite transport, and the minimum-action geometry introduced later.

VIII. CONNECTED SPECTRAL CUMULANTS AND A HYPERCYCLE THEOREM

The correct mathematical language for non-additivity is not “pairwise synergy plus patches.” It is the calculus of *connected mixed derivatives*. The logarithm of a determinant plays the role of a connected generating functional: mixed derivatives remove products of disconnected effects in exactly the same way that cumulants remove disconnected moments.

A. Real spectral potential and connected action derivatives

At a fixed s away from the spectrum define the branch-free spectral potential

$$\Psi(s, a) = \log |\det T(s, a)|. \quad (38)$$

For a nonempty action set $S = \{i_1, \dots, i_k\}$, define the connected spectral cumulant density

$$\kappa_S(s, a) = D_{i_1} \cdots D_{i_k} \Psi(s, a). \quad (39)$$

For complex phase-sensitive bookkeeping one may work locally with $D_S \log \det T$; all cooperative-game quantities used for engineering allocation are kept real by using (38) or another real security objective.

The first two derivatives are

$$D_i \log \det T = \text{tr}(G D_i T), \quad (40)$$

$$D_{ij} \log \det T = \text{tr}(G D_{ij} T - G D_i T G D_j T), \quad (41)$$

where $G = T^{-1}$. Equation (41) exposes two physically distinct sources of non-additivity: $G D_{ij} T$ is an intrinsic/re-equilibration interaction vertex, whereas $-G D_i T G D_j T$ is a feedback-mediated closed return through the network-controller resolvent.

Theorem 4 (Connected partition-cycle formula). *Let S be a finite set of action indices and let $\Pi(S)$ denote its set partitions. For a block $B \subseteq S$, write $T_B = D_B T$. Then*

$$D_S \log \det T = \sum_{\pi \in \Pi(S)} \frac{(-1)^{|\pi|-1}}{|\pi|} \sum_{\sigma \in \mathfrak{S}_{|\pi|}} \text{tr} \left[\prod_{r=1}^{|\pi|} G T_{B_{\sigma(r)}} \right], \quad (42)$$

where $\pi = \{B_1, \dots, B_{|\pi|}\}$. *Cyclically equivalent products have identical trace and may be grouped computationally.*

Proof. Apply the multivariate chain rule to $\log \det T = \text{tr} \log T$ and use $D_i G = -G D_i T G$. Group the resulting terms by the partition of S that records which action derivatives land on the same derivative block $D_B T$. For a partition with r blocks, differentiation of the matrix logarithm generates all $r!$ orderings with coefficient $(-1)^{r-1}/r$. For $|S| = 2$ this yields (41); for $|S| = 3$ it gives one $G T_{123}$ term, three negative two-block terms, and the two cyclically inequivalent positive three-return traces. Induction on $|S|$ completes the formula. $\square$

Theorem 4 is the mathematical replacement for an ad hoc collection of pair/triple formulas. It also reveals a new physical object. Each T_B is a possibly multi-action *interaction vertex*; each G transports that vertex through the entire grid; and the trace closes the transported blocks into a connected hypercycle.

Re-equilibration therefore does not break the cycle theory—it generalizes an ordinary action quiver into a partitioned interaction hypergraph.

B. Affine fixed-equilibrium limit: cycles emerge exactly

The common low-rank construction is recovered as a corollary.

Corollary 5 (Affine connected-cycle theorem). *Suppose locally*

$$T(s, a) = T_0(s) + \sum_{i=1}^m a_i B_i(s), \quad X_i = G_0 B_i. \quad (43)$$

For k distinct actions $S = \{i_1, \dots, i_k\}$,

$$D_S \log \det T|_{a=0} = (-1)^{k-1} \sum_{\pi \in \mathfrak{S}_k / C_k} \text{tr}(X_{\pi_1} \cdots X_{\pi_k}). \quad (44)$$

Thus the genuine k -action connected term is the sum of oriented closed returns that visit every selected action exactly once.

Proof. In (42), all T_B with $|B| > 1$ vanish for an affine pencil. Only the singleton partition remains. Dividing the $k!$ permutations by the k cyclic rotations that have identical trace gives the quotient $\mathfrak{S}_k / C_k$. $\square$

If $B_i = U_i C_i V_i^*$ is low rank, every factor in (44) can be contracted to port space:

$$V_i^* G_0 U_j. \quad (45)$$

This is the origin of the action quiver. The quiver is therefore an efficient realization of the connected calculus when the tangent actions are local/low rank; it is not an assumption imposed on a finite re-equilibrated change.

C. Finite externalities are integrals of connected densities

Finite coalition effects admit an exact bridge to the local cumulants.

Theorem 6 (Finite dividend-cumulant identity). *Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be $C^{|S|}$ on the action hyperrectangle $0 \leq a_i \leq \alpha_i$. Define the anchored Harsanyi/Möbius dividend*

$$\Delta_S f = \sum_{U \subseteq S} (-1)^{|S|-|U|} f(\alpha_U), \quad (46)$$

where $(\alpha_U)_i = \alpha_i$ for $i \in U$ and zero otherwise. Then

$$\Delta_S f = \int_0^{\alpha_{i_1}} \cdots \int_0^{\alpha_{i_k}} D_S f(a_S) da_{i_k} \cdots da_{i_1}. \quad (47)$$

Proof. Apply the one-dimensional fundamental theorem of calculus successively in each coordinate. Each integration contributes a forward finite difference; the product of the k differences is exactly the Boolean-lattice Möbius sum (46). $\square$

With $f = \Psi(s, a)$, Theorem 6 states that a finite spectral externality is the integral of the connected re-equilibrated interaction density over the entire action cell. With $f = \zeta_{\min}(a)$ or an EMT violation metric, the same identity defines an exact real engineering dividend whenever the objective is smooth enough on the cell. Shapley allocation is then applied to these

real finite dividends; no multivalued complex logarithm is required.

This creates a useful two-lattice picture. The partition lattice in Theorem 4 removes disconnected differential contributions; the Boolean subset lattice in (46) removes lower-order finite coalition contributions. Möbius inversion is a general language for higher-order interactions across many fields [73]; here the two structures are connected by (47) and anchored to the physical AC equilibrium manifold.

D. Exact finite hypercycle decomposition

The local partition theorem and the finite cubical identity combine into a stronger endpoint result: the finite portfolio externality itself can be decomposed by interaction topology, without freezing the operating point.

Theorem 7 (Finite connected-hypercycle decomposition). *Let S be a finite action set and assume $T(s, a)$ remains nonsingular on the action cell $\mathcal{Q}_S$. For $\Psi = \log |\det T|$, define for each set partition $\pi = \{B_1, \dots, B_r\} \in \Pi(S)$*

$$\mathcal{E}_\pi(S; s) = \int_{\mathcal{Q}_S} \Re \left\{ \frac{(-1)^{r-1}}{r} \sum_{\sigma \in \mathfrak{S}_r} \text{tr} \left[\prod_{q=1}^r GT_{B_{\sigma(q)}} \right] \right\} da_S. \quad (48)$$

Then the exact finite spectral externality satisfies

$$\Delta_S \Psi = \sum_{\pi \in \Pi(S)} \mathcal{E}_\pi(S; s). \quad (49)$$

Proof. Insert Theorem 4 into the finite dividend identity (47), take real parts, and interchange the finite partition/permutation sums with the integral. $\square$

The partition size r has a direct mechanism interpretation. The one-block partition $\{S\}$ is a pure multi-action equilibrium/structural vertex. The all-singleton partition has $r = |S|$ and is the ordinary resolvent-mediated action cycle. Intermediate partitions are *hybrid hypercycles*: some actions first combine through the nonlinear re-equilibrated pencil and the resulting vertices then close through network/controller transport. This is the exact finite version of the paper's central reconstruction.

For two actions,

$$\begin{aligned} \Delta_{ij} \Psi = & \underbrace{\int_{\mathcal{Q}_{ij}} \Re \text{tr}(GT_{ij}) da_i da_j}_{\mathcal{E}_{\{ij\}}: \text{intrinsic}} \\ & - \underbrace{\int_{\mathcal{Q}_{ij}} \Re \text{tr}(GT_i GT_j) da_i da_j}_{-\mathcal{E}_{\{i\}\{j\}}: \text{closed feedback}}. \end{aligned} \quad (50)$$

Thus a sign reversal can be audited rather than merely observed: it may arise because the equilibrium manifold curves strongly, because a controller-mediated return is destructive, or because both terms change with the finite action level.

For three actions, the exact dividend contains one three-action vertex, three pair-plus-single hypercycles, and two oriented pure three-action returns. The same partition hierarchy extends to

arbitrary order and avoids the false dichotomy “either operating-point effect or cycle effect.” The theory measures both in one connected decomposition.

E. A certificate for when the ordinary action quiver is adequate

The low-rank/quiver model keeps only the all-singleton partition. Theorem 7 therefore yields a direct error certificate for that approximation.

Let $k = |S|$, let $\pi_0 = \{\{i\} : i \in S\}$, and denote the all-singleton cycle contribution by κ_S^{cyc} . The omitted re-equilibrium curvature is

$$R_S^{\text{curv}} = D_S \log \det T - \kappa_S^{\text{cyc}}. \quad (51)$$

If the dynamic pencil has dimension n_T , then

$$|R_S^{\text{curv}}| \leq n_T \sum_{\substack{\pi \in \Pi(S) \\ \pi \neq \pi_0}} (|\pi| - 1)! \prod_{B \in \pi} \|GT_B\|_2. \quad (52)$$

Indeed, the coefficient $1/|\pi|$ in (42) multiplies $|\pi|!$ orderings, and $|\text{tr}(A_1 \cdots A_r)| \leq n_T \prod_q \|A_q\|_2$.

Integrating (52) over $\mathcal{Q}_S$ bounds the error made when the finite dividend is approximated using only ordinary action cycles. This produces a principled *quiver adequacy gate*: if the integrated curvature remainder is below the decision tolerance, the action quiver is a certified simplification; otherwise the higher interaction vertices $D_B T$ must be retained. In particular, the framework never needs to pretend that re-solving the AC operating point preserves a globally low-rank finite update.

F. Controlled numerical identity check

A four-state nonlinear toy model with a scalar re-equilibrium variable was used solely to audit the identities. For the tested three-action finite move, freezing the equilibrium and retaining only the structural affine update predicted $\Delta \Psi = -1.2745 \times 10^{-2}$, whereas the re-equilibrated operator gave $+5.8416 \times 10^{-3}$. The exact path integral (35) matched the finite determinant change to 5.7×10^{-16} . The three-action finite dividend was 3.67563×10^{-5} ; numerical integration of the mixed third derivative in (47) differed by 1.35×10^{-10} . The point of the example is not its physical realism; it demonstrates why the re-equilibrated theory is not a perturbative patch to the frozen model.

IX. FINITE DETERMINANTS, CYCLE COVERS, AND CONNECTED CYCLE CLUSTERS

The previous section identifies connected differential interactions. When a low-dimensional action-quiver representation is available, determinant combinatorics provide a second exact description. This manuscript separates three objects that were previously conflated: simple cycles, primitive cyclic walks, and connected clusters of overlapping cycles.

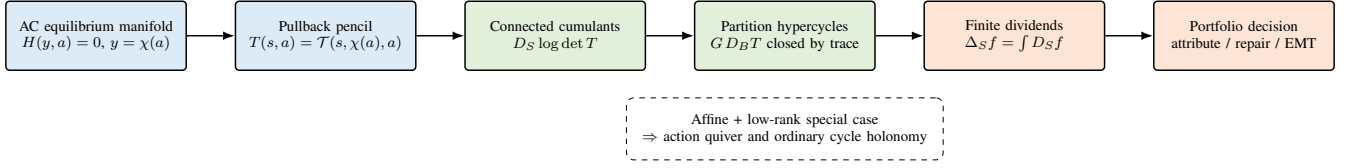


Fig. 1. Reconstructed mathematical backbone. Re-equilibrated actions are handled exactly first. Ordinary controller-mediated action cycles emerge as the affine/low-rank realization of a more general connected hypercycle calculus rather than being imposed as a universal finite-rank assumption.

A. Normalized pure-interaction operator

For an affine low-rank action model, collect the port update as

$$T_A = T_0 + U_A C_A V_A^*, \quad K_A = C_A V_A^* G_0 U_A. \quad (53)$$

The determinant lemma gives

$$\frac{\det T_A}{\det T_0} = \det(I + K_A). \quad (54)$$

Let $D = \text{blkdiag}(I + K_{11}, \dots, I + K_{mm})$ and define the zero-diagonal block interaction matrix Q by

$$I + K = D(I + Q). \quad (55)$$

Then

$$\det(I + K) = \prod_i \det(I + K_{ii}) \det(I + Q). \quad (56)$$

The first factor is the isolated action physics at the linearization point; $\det(I + Q)$ contains only cross-action composition.

Theorem 8 (Acyclic composability and SCC factorization). *If the directed support of Q is acyclic, $\det(I + Q) = 1$. More generally, if Q_c are the diagonal blocks after permutation to SCC condensation form,*

$$\det(I + Q) = \prod_c \det(I + Q_c). \quad (57)$$

Proof. Acyclic support admits a topological ordering making Q strictly block triangular. SCC condensation makes $I + Q$ block triangular with diagonal blocks $I + Q_c$. The determinant statements follow immediately. $\square$

The theorem is exact but does not guarantee that a raw resolvent-generated graph is sparse. Section XI therefore derives a principled localization layer rather than thresholding edges arbitrarily.

B. Exact finite cycle-cover expansion

The determinant itself has a finite combinatorial representation that requires no convergence assumption.

Theorem 9 (Cycle-system expansion). *For scalar zero-diagonal Q , $\det(I + Q)$ equals one plus the signed sum over all collections $\mathcal{C}$ of pairwise vertex-disjoint directed simple cycles:*

$$\det(I + Q) = 1 + \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-1)^{|C|-1} w(C), \quad (58)$$

where $w(C) = \prod_{(j \rightarrow i) \in C} q_{ij}$. Fixed points of the underlying permutation contribute the unit diagonal.

Proof. Apply the Leibniz determinant formula to $I + Q$. Every permutation decomposes uniquely into disjoint cycles; its fixed points select diagonal ones, while each nontrivial cycle selects the corresponding directed entries of Q . The sign of a length- ℓ cycle is $(-1)^{\ell-1}$. Summing over permutations yields (58); this is the standard cycle-cover interpretation of determinants [74]. $\square$

Theorem 9 explains why an isolated chordless cycle has a one-factor determinant, but it also shows a limitation: overlapping simple cycles cannot simultaneously appear in the same cycle cover. Their collective effect is encoded by the logarithm/connected-cluster expansion rather than by multiplying naive simple-cycle factors.

C. Closed walks, prime cyclic walks, and cluster expansion

When $\rho(Q) < 1$,

$$\log \det(I + Q) = \sum_{p \geq 1} \frac{(-1)^{p+1}}{p} \text{tr}(Q^p). \quad (59)$$

The traces enumerate closed walks of length p , including repeated vertices and repeated traversals. Kassel and Lévy provide a recent block/quiver determinant theory formulated in terms of traces of holonomies and prime cyclic walks [23]. The terminology matters: an exact Euler-type product, when invoked, is over the appropriate primitive cyclic-walk classes, not merely over graph-theoretic simple cycles.

For engineering interpretation it is often preferable to retain simple cycles but organize their overlaps. Define a cycle-overlap graph whose vertices are simple action cycles and whose edges join cycles sharing at least one action. Connected components of this overlap graph are *cycle clusters*. The logarithm is the natural connected object: in statistical mechanics the linked-cluster/Mayer expansion similarly expresses the logarithm of a partition function through connected clusters, with Kotecký–Preiss-type criteria controlling convergence [71], [72]. This analogy suggests a rigorous scalable approximation: enumerate only connected action-cycle clusters whose certified tail is below the planning tolerance.

The present paper therefore uses the following hierarchy:

$$\begin{array}{c} \det(I + Q) : \text{finite disjoint cycle systems} \\ \Downarrow \\ \log \det(I + Q) : \text{connected/repeated cycle clusters} \end{array} \quad (60)$$

This removes the earlier ambiguity between “prime cycle” and “simple cycle” and gives overlapping cycles a mathematically natural home.

D. Exact polymer representation and linked cycle clusters

The cycle-system determinant admits an exact reformulation as a hard-core polymer partition function. Let $\mathcal{C}$ be the set of directed simple cycles of the scalar action graph and assign each cycle the activity

$$z_C = (-1)^{|C|-1} w(C). \quad (61)$$

Two cycles are compatible, written $C \sim C'$, if they are vertex disjoint. Theorem 9 is then exactly

$$Z(Q) := \det(I + Q) = \sum_{\Gamma \text{ compatible}} \prod_{C \in \Gamma} z_C, \quad (62)$$

where the empty family contributes one. This is not an analogy: $Z(Q)$ is literally the partition function of a hard-core gas whose polymers are action cycles and whose incompatibility is cycle overlap.

The standard Mayer/Ursell linked-cluster expansion therefore gives an exact connected organization of $\log Z$ whenever the expansion converges. For an ordered n -tuple $\mathbf{C} = (C_1, \dots, C_n)$ define

$$f(C_i, C_j) = \begin{cases} -1, & C_i \not\sim C_j, \\ 0, & C_i \sim C_j, \end{cases} \quad (63)$$

and the Ursell coefficient

$$\phi^T(\mathbf{C}) = \sum_{G \in \mathcal{G}_n^{\text{conn}}} \prod_{(i,j) \in E(G)} f(C_i, C_j), \quad (64)$$

where the sum is over connected simple graphs on $\{1, \dots, n\}$. Then

$$\log \det(I + Q) = \sum_{n \geq 1} \frac{1}{n!} \sum_{C_1, \dots, C_n \in \mathcal{C}} \phi^T(C_1, \dots, C_n) \times \prod_{r=1}^n z_{C_r}. \quad (65)$$

This is the rigorous form of the phrase “connected overlapping-cycle clusters.” Only tuples whose incompatibility graph is connected survive in the logarithm. Disjoint cycle families belong to the determinant itself and cancel from the connected logarithm.

A practical sufficient convergence gate can be imported from abstract polymer theory [71], [72]. If there exists a nonnegative weight $a(C)$ such that

$$\sum_{C': C' \not\sim C} |z_{C'}| e^{a(C')} \leq a(C) \quad \text{for every } C, \quad (66)$$

then the linked-cluster series converges absolutely. Equation (66) is not a power-system stability criterion; it is a certificate that truncating the cycle-cluster representation is mathematically controlled. When it fails, the exact SCC determinant remains the fallback.

This representation yields a useful hierarchy of interaction descriptions. A single isolated cycle is the one-polymer term. Two overlapping cycles first interact through their two-polymer Ursell coefficient. Larger connected overlap complexes provide a systematic definition of higher-order *cycle-cluster externality*. The three-action counterexample in Appendix D demonstrates why this level is necessary: two individually nonclosing cycles can close collectively through their shared action.

E. Relationship to prime cyclic walks

The polymer expansion and the prime-cyclic-walk expansion answer different combinatorial questions and should not be conflated. Equation (65) starts from graph-theoretic simple cycles and accounts for overlap by connected clusters. The trace/quiver expansion starts from closed walks, including repeats, and can be reorganized into primitive cyclic-walk classes as in the determinant identities of Kassel and Lévy [23]. The former is natural for naming finite engineering mechanisms; the latter is natural for exact algebraic resummation. Agreement of the two descriptions provides a valuable implementation cross-check on small action cores.

X. MIMO ACTION QUIVERS, HOLONOMY ALGEBRA, AND SCALARIZABILITY

Power-electronic ports are generally MIMO. A PLL/current-controller/network interface cannot be reduced a priori to one scalar edge without potentially destroying directional information. Let action i carry a port space $\mathcal{V}_i \simeq \mathbb{C}^{r_i}$ and let $Q_{ij}(s) : \mathcal{V}_j \rightarrow \mathcal{V}_i$ be the normalized transport block. The collection $(\mathcal{V}_i, Q_{ij})$ is a frequency-dependent quiver representation.

A. Cycle holonomy

For an oriented cycle $C = (i_1, \dots, i_\ell)$ define the ordered return

$$\mathcal{H}_C = Q_{i_1 i_\ell} Q_{i_\ell i_{\ell-1}} \cdots Q_{i_2 i_1}. \quad (67)$$

Under local changes of port basis $Q_{ij} \mapsto S_i^{-1} Q_{ij} S_j$,

$$\mathcal{H}_C \mapsto S_{i_1}^{-1} \mathcal{H}_C S_{i_1}. \quad (68)$$

Hence the characteristic polynomial, spectrum, trace, and determinant of $\mathcal{H}_C$ are coordinate invariants. Singular-value distances are invariant only under unitary/physically normalized coordinate changes; this manuscript does not label a raw Frobenius commutator magnitude as a general gauge invariant.

Theorem 10 (Chordless block-cycle closure). *If an SCC contains exactly one chordless directed block cycle and zero diagonal blocks, then*

$$\det(I + Q_C) = \det[I - (-1)^{|C|} \mathcal{H}_C]. \quad (69)$$

Thus the isolated cycle is singular if and only if $(-1)^{|C|} \mathcal{H}_C$ has eigenvalue one.

Proof. Block elimination around the cycle expresses each variable in terms of the previous one. After one traversal, a nonzero initial vector must satisfy $x = (-1)^{|C|} \mathcal{H}_C x$. Unit block pivots preserve the determinant, giving (69). $\square$

For overlapping cycles, Theorem 10 is not used as a universal criterion. The SCC-level quantity

$$\Gamma_S(s) = \|(I + Q_S(s))^{-1}\|_2 \quad (70)$$

retains collective interference, while individual holonomies remain mechanism candidates that can be tested by cycle surgery.

B. Holonomy algebra rather than an ad hoc non-Abelian score

Choose a basis of fundamental cycles in one SCC and define the generated holonomy algebra

$$\mathcal{A}_H(s) = \text{alg}\{I, \mathcal{H}_{C_1}(s), \dots, \mathcal{H}_{C_q}(s)\}. \quad (71)$$

Its commutant is

$$\mathcal{C}_H(s) = \{X : XH = HX \text{ for all } H \in \mathcal{A}_H(s)\}. \quad (72)$$

The algebraic question “can the interaction be represented by independent scalar channels?” is now precise.

Proposition 11 (Exact simultaneous scalarization criterion). *If the fundamental holonomies are diagonalizable, they admit one common scalar-channel basis if and only if they commute pairwise. More generally, nontrivial invariant subspaces of $\mathcal{A}_H$ define the finest common block decomposition; the commutant (72) can be used to compute it.*

Proof. A commuting family of diagonalizable complex matrices is simultaneously diagonalizable. Conversely, matrices diagonal in one basis commute. The block statement follows from standard finite-dimensional representation theory of matrix algebras. $\square$

This proposition deliberately claims no novelty for the linear-algebra theorem. The power-system contribution is the construction of the holonomy algebra from controller/network action ports and the use of its reducibility as a diagnostic for when a scalar strength representation is structurally defensible.

XI. LOCALITY, CERTIFIED SPARSIFICATION, AND SCALABLE FEEDBACK CORES

A resolvent is usually dense, so an action quiver constructed after aggressive Kron reduction can collapse into one giant SCC. Scalability therefore requires a structural reason for discarding weak remote couplings. The correct starting point is the sparse local descriptor representation before global Kron reduction.

A. Resolvent locality

For sparse diagonalizable matrices and analytic matrix functions, entries of $f(A)$ often decay with graph distance under spectral-separation conditions; Benzi and Razouk establish such bounds and corresponding linear-complexity approximation strategies for sparse matrix functions [70]. Resolvents are matrix functions, so away from poles one expects a bound of the form

$$\|V_i^* G(s) U_j\| \leq C(s) \exp[-\alpha(s)d(i, j)] \quad (73)$$

for local port supports in regimes where the hypotheses of the matrix-decay theory hold. Near a lightly damped global mode, $\alpha(s)$ can collapse; that loss of locality is itself diagnostically important and should force a larger core rather than be hidden by a fixed graph radius.

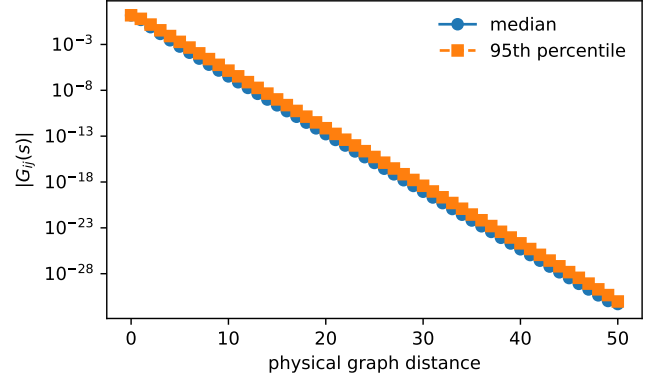


Fig. 2. Controlled sparse-operator check: resolvent transport decays approximately exponentially with physical graph distance away from a pole. The plot is a mathematical sanity check, not an IEEE-system claim.

B. A polynomial-approximation locality theorem

The exponential form in (73) can be given a direct sufficient-condition proof that also exposes when locality should fail. Let $\mathcal{G}(T)$ be the sparsity graph of a square local operator T and let $d(i, j)$ be graph distance between index blocks. If p_k is any polynomial of degree at most $k < d(i, j)$, then

$$[p_k(T)]_{ij} = 0, \quad (74)$$

because no product of at most k local couplings can connect the two supports. Consequently,

$$\|[T^{-1}]_{ij}\| \leq \inf_{p \in \mathcal{P}_{d(i,j)-1}} \|T^{-1} - p(T)\|_2. \quad (75)$$

Proposition 12 (Spectral-separation locality). *Assume T is diagonalizable, $T = V\Lambda V^{-1}$, and its spectrum lies in a compact set $K \subset \mathbb{C}$ with $0 \notin K$. Then*

$$\|[T^{-1}]_{ij}\| \leq \kappa_2(V) E_{d(i,j)-1}(K), \quad (76)$$

where

$$E_k(K) = \inf_{p \in \mathcal{P}_k} \max_{z \in K} |z^{-1} - p(z)|. \quad (77)$$

Whenever $1/z$ admits geometrically convergent polynomial approximation on K , $E_k(K) \leq Cq^k$ with $q < 1$, producing exponential decay with graph distance. For normal T , $\kappa_2(V) = 1$.

Proof. Equation (74) gives $[T^{-1}]_{ij} = [T^{-1} - p(T)]_{ij}$ for every admissible p . For a diagonalizable matrix,

$$\|T^{-1} - p(T)\|_2 \leq \kappa_2(V) \max_{\lambda \in \text{spec } T} |\lambda^{-1} - p(\lambda)|. \quad (78)$$

Minimizing over p and enlarging the maximum from $\text{spec } T$ to K proves (76). $\square$

Proposition 12 supplies the missing logic behind action-graph sparsification. Locality degrades for two mathematically distinct reasons: the spectral set approaches zero, worsening the approximation of $1/z$, or the eigenbasis becomes ill conditioned, increasing $\kappa_2(V)$. The latter is precisely a nonnormality penalty. A controller-rich grid can therefore lose action locality before it becomes unstable, in which case a giant SCC is a physical

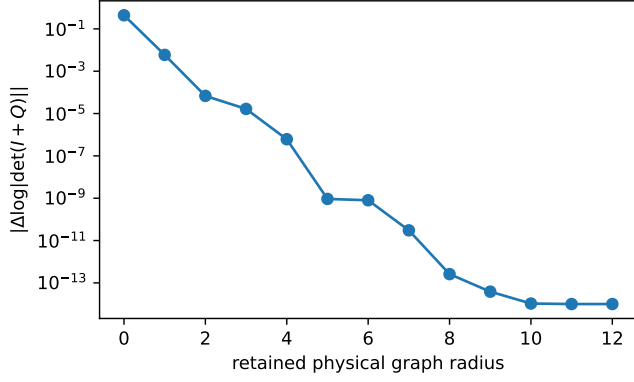


Fig. 3. Controlled 40-action sparsification check. Retaining only interactions within a modest physical radius rapidly recovers the exact log interaction determinant for a sparse local operator. A real grid must verify the decay/certificate frequency by frequency rather than assume locality.

warning rather than an algorithmic nuisance. The general sparse-matrix-function decay theory of Benzi and Razouk provides sharper variants for broader matrix classes [70].

For local action supports U_j, V_i , the same bound applies to the transport block:

$$\|V_i^* T^{-1} U_j\|_2 \leq \|V_i\|_2 \|U_j\|_2 \kappa_2(V) E_{d(i,j)-1}(K). \quad (79)$$

This turns physical graph distance, spectral separation, and nonnormality into one sufficient pruning certificate rather than treating distance as a universal proxy for interaction strength.

C. Certified pruning

Let Q be the exact action interaction matrix and $\tilde{Q} = Q - E$ a sparsified approximation. Define

$$\eta = \|(I + \tilde{Q})^{-1} E\|_2. \quad (80)$$

If $\eta < 1$, then $I + Q$ is nonsingular by the Banach perturbation lemma. If r is the action-core dimension, the branch-free determinant error satisfies

$$\left| \log |\det(I + Q)| - \log |\det(I + \tilde{Q})| \right| \leq -r \log(1 - \eta). \quad (81)$$

Appendix F proves the bound. In computation, the exact determinant ratio can also be evaluated in the small pruned remainder.

The algorithmic consequence is important: SCC localization becomes credible only after a frequency-dependent certified pruning step. A remote device remains in the core whenever its Green transport violates the tolerance, regardless of geographic distance.

XII. COMPLEX FREQUENCY AS A LOCAL TEMPORAL JET, NOT A GLOBAL INTERACTION SURROGATE

Milano and co-workers provide a complementary local-physics layer. Complex frequency (CF) combines relative voltage-magnitude variation and angular-frequency motion, while the recent power-system-strength framework organizes resistance of voltage, CF, and CF rate to sudden current changes into zero-, first-, and second-order indicators [45], [46]. The

present paper uses these results as baselines and observable coordinates rather than redefining them.

For a nonzero complex voltage $v(t)$,

$$\eta_v = \frac{\dot{v}}{v} = \rho_v + j\omega_v. \quad (82)$$

At a steady operating point v_0 , a first-order perturbation gives

$$\delta\eta_v(s) = s D_{v0}^{-1} C_v \delta x(s), \quad (83)$$

so the full resolvent projects directly into a CF observation map

$$H_{\eta a}(s) = s D_{v0}^{-1} C_v G(s) U_A. \quad (84)$$

The conceptual distinction can be expressed as two independent derivative axes:

$$\begin{array}{l} J_t^{(0;2)} : \text{local temporal response / CF strength,} \\ D_a^k \Psi : \text{connected multi-action interaction.} \end{array} \quad (85)$$

The first asks how a device/bus responds in time to an injection jump; the second asks how several engineering actions fail to compose after the full equilibrium and network-controller feedback are accounted for. A strong local temporal jet can coexist with a large connected action cumulant. This is the precise version of the manuscript's central phrase: *a strong grid can contain a weak dynamic interaction core*.

Sofos and Milano show that CF can coincide with eigenvalues in appropriate transformed coordinates for diagonalizable LTI systems and that geometric frequency remains meaningful more broadly along nonlinear flows [21]. That result provides a physical interpretation of the spectral coordinate; the connected calculus developed here provides a mechanism decomposition for how finite engineering actions move that coordinate. A stronger exact mapping between the twelve strength indicators and particular moments/projections of $G(s)$ is left as a theorem target because it requires reproducing the Delta-operator treatment of algebraic jumps exactly rather than equating the indicators loosely with Markov parameters.

XIII. BAND-INTEGRATED REAL SPECTRAL POTENTIALS FOR PLANNING AND ATTRIBUTION

A single complex frequency is useful for mechanism diagnosis but is an awkward basis for portfolio ranking: the choice of s can be arbitrary, $\log \det$ is branch sensitive, and a lightly damped mode can move across the sampled grid. The connected calculus can instead be applied to a real frequency-band functional.

For a nonnegative weight $w(\omega)$ and regularization $\varepsilon > 0$, define

$$\Phi_{\Omega, \varepsilon}(a) = \frac{1}{2} \int_{\Omega} w(\omega) \log \det [T(j\omega, a)^* T(j\omega, a) + \varepsilon^2 I] d\omega. \quad (86)$$

The integrand is real and smooth. For a square nonsingular T , as $\varepsilon \downarrow 0$,

$$\frac{1}{2} \log \det (T^* T) = \log |\det T|, \quad (87)$$

so (86) is a regularized band-integral of the branch-free spectral potential used earlier. The weight can emphasize a known PLL,

inter-area, or protection-relevant band without defining a new universal stability index.

Because differentiation and integration commute under the usual dominated-smoothness assumptions,

$$D_S \Phi_{\Omega, \varepsilon} = \int_{\Omega} w(\omega) D_S \psi_{\varepsilon}(\omega, a) d\omega, \quad (88)$$

where ψ_{ε} is the integrand in (86). The finite dividend identity then gives

$$\Delta_S \Phi_{\Omega, \varepsilon} = \int_{Q_S} \int_{\Omega} w(\omega) D_S \psi_{\varepsilon}(\omega, a) d\omega da_S. \quad (89)$$

Thus a project-set attribution can be both finite in action amplitude and integrated over a physically selected frequency band, with no complex-log branch choice.

The regularization parameter has a clear interpretation. Far from singularity, ε has negligible effect. Near a pole/zero, it caps the logarithmic barrier and improves numerical conditioning. A sensitivity analysis in ε must accompany any reported allocation; the true stability decision remains tied to eigenvalue/contour/EMT criteria, while (86) is an attribution and ranking potential.

A. Relation to resolvent amplification

The band potential and the SCC interaction gain answer complementary questions. $\Phi_{\Omega, \varepsilon}$ summarizes how the singular-value volume of the dynamic operator changes, whereas

$$\Gamma_S(\omega) = \|(I + Q_S(j\omega))^{-1}\|_2 \quad (90)$$

tracks the worst interaction direction in a connected action core. A narrow singular direction may produce a large Γ_S with a moderate log-volume change; a broad degradation can produce a large band dividend without one extreme peak. Reporting both avoids conflating worst-case amplification with aggregate spectral deformation.

XIV. MECHANISTIC LOCALIZATION: WAVEMAKERS, ENDOGENEITY, AND INTERACTION RESOLVENTS

The location where an oscillation is visible need not be the location that creates it. Fluid-mechanics global-stability analysis distinguishes direct-mode amplitude, adjoint receptivity, wavemaker/structural sensitivity, and endogeneity [56], [57]. The same separation is useful here.

Consider a generalized eigenpair

$$Ar = \lambda Er, \quad l^* A = \lambda l^* E, \quad l^* Er = 1. \quad (91)$$

If $A = \sum_q A_q$ is a physical device/network partition, then

$$\lambda = \sum_q \varepsilon_q, \quad \varepsilon_q = l^* A_q r. \quad (92)$$

This exact budget separates which modeled blocks contribute to growth/decay and frequency. For an admissible infinitesimal port feedback $\delta A_i = B_i \Delta_i C_i$, $\|\Delta_i\|_F \leq 1$,

$$W_i = \max |\delta \lambda_i| = \|B_i^* l\|_2 \|C_i r\|_2. \quad (93)$$

Thus right-mode visibility, adjoint receptivity, wavemaker leverage, and endogeneity answer four distinct questions.

The action calculus adds a fifth: *what closes?* For one SCC, Γ_S from (70) measures collective interaction amplification. For a credible isolated cycle candidate, the eigenvalues of $(-1)^{|C|} \mathcal{H}_C$ indicate proximity to one-traversal closure. A mechanism claim is strengthened by *cycle surgery*: perturb an action on the identified cycle, predict the modal change, verify it in the full model, and compare against a matched perturbation outside the cycle. The paper therefore uses the phrase *mechanistic/intervention-based attribution*, reserving stronger causal language for experiments with such negative controls.

XV. STRUCTURED ACTION GEOMETRY: DISTANCE TO FAILURE AND MINIMUM RECOVERY

Let $\mathcal{U}$ denote the unsafe set in action space, defined by a spectral, damping, waveform, protection, or EMT criterion. Let $W(a)$ encode engineering scale/cost. Two distances must be distinguished.

$$r^-(a_0) = \inf_{\Delta a} \{\|W \Delta a\|_2 : a_0 + \Delta a \in \mathcal{U}\}, \quad (94)$$

$$r^+(a_u; \delta) = \inf_{\Delta a} \{\|W \Delta a\|_2 : \mathcal{M}(a_u + \Delta a) \geq \delta\}, \quad (95)$$

where $\mathcal{M}$ is the chosen positive safety margin. r^- is an adversarial distance-to-failure; r^+ is a minimum recovery cost. They are dual design questions but not the same optimization problem.

For a smooth simple eigenvalue boundary, (37) supplies local normals and first-order initializations. For a named isolated cycle, one may use an invariant spectral distance

$$d_C^{\text{eig}}(s) = \min_{\nu \in \text{spec}((-1)^{|C|} \mathcal{H}_C(s))} |1 - \nu| \quad (96)$$

under arbitrary local similarities. In physically normalized port coordinates, the singular-value distance in Theorem 10 can additionally be used for robust perturbation bounds. For overlapping-cycle cores, $r^{\pm}$ are defined directly on $\det(I + Q_S)$ or on the full physical security metric rather than forcing one cycle to explain a collective closure.

A. Contour reserve and a certified lower bound on action distance

The Gohberg–Sigal/Rouché condition also yields a useful security reserve before solving the nonlinear minimum-action problem. Define the contour conditioning reserve

$$\delta_{\Gamma}(a) = \inf_{s \in \Gamma} \sigma_{\min}(T(s, a)) = \left[\sup_{s \in \Gamma} \|T^{-1}(s, a)\|_2 \right]^{-1}. \quad (97)$$

A small δ_{Γ} is a structured-pseudospectral warning: the system can be close to a contour crossing even when the nearest nominal eigenvalue appears farther away in the Euclidean s plane.

Proposition 13 (Certified action-radius lower bound). *Suppose that in a weighted action ball $\|W \Delta a\|_2 \leq r$ the re-equilibrated pencil satisfies the finite Lipschitz bound*

$$\sup_{s \in \Gamma} \|T(s, a + \Delta a) - T(s, a)\|_2 \leq L_{\Gamma} \|W \Delta a\|_2. \quad (98)$$

Then no action with

$$\|W\Delta a\|_2 < \frac{\delta_\Gamma(a)}{L_\Gamma} \quad (99)$$

can change the zero/mode count across Γ . Consequently,

$$r^-(a) \geq \delta_\Gamma(a)/L_\Gamma. \quad (100)$$

Proof. For every $s \in \Gamma$,

$$\|T^{-1}\Delta T\|_2 \leq \|T^{-1}\|_2 L_\Gamma \|W\Delta a\|_2 < 1. \quad (101)$$

The matrix Rouché/Gohberg–Sigal condition therefore preserves the contour count. $\square$

A computable conservative L_Γ follows from the total derivative:

$$L_\Gamma \leq \sup_{\substack{s \in \Gamma \\ a' \in \mathcal{B}}} \|D_a T(s, a') W^{-1}\|_{2 \leftarrow 2}, \quad (102)$$

where $\mathcal{B}$ is the certified action neighborhood. Equation (100) supplies a guaranteed safe radius; the optimization defining r^- searches for the actual nearest failure. The ratio between the two quantifies conservatism and provides a natural benchmark for structured-pseudospectral refinements.

B. Second-order geometry of a smooth stability boundary

If a scalar margin $m(a)$ parameterizes a smooth piece of the action discriminant, its weighted normal is $g^{-1}\nabla m$ with $g = W^T W$. The Hessian projected onto the tangent space gives the second fundamental form of the boundary. This matters because a first-order sensitivity can rank actions correctly yet badly estimate a finite safe region when the boundary is strongly curved. The connected second derivative (41) supplies exactly the terms responsible for that curvature: an intrinsic equilibrium contribution and a closed feedback contribution. Thus boundary geometry, portfolio externality, and controller-mediated cycles are not separate add-ons; they are different projections of the same pullback derivatives.

XVI. CYCLE-PRESERVING MODEL FIDELITY AND CERTIFIED EMT REGION SELECTION

The most consequential power-system application is not “replace EMT by holonomy.” It is to select the smallest detailed region whose external equivalent preserves the dangerous dynamic mechanism. This problem is naturally expressed by the same Schur self-energy that started the paper.

A. Exact exterior self-energy

Partition a detailed dynamic operator into a candidate core c and exterior e :

$$T(s) = \begin{bmatrix} T_{cc} & T_{ce} \\ T_{ec} & T_{ee} \end{bmatrix}. \quad (103)$$

Eliminating the exterior gives the exact core operator

$$T_{\text{eff},c} = T_{cc} - \Sigma_e(s), \quad \Sigma_e = T_{ce} T_{ee}^{-1} T_{ec}. \quad (104)$$

If a proposed RMS/phasor/reduced exterior supplies $\hat{\Sigma}_e^R$, the only model mismatch seen by the retained core is

$$\Delta \Sigma_R(s) = \Sigma_e(s) - \hat{\Sigma}_e^R(s). \quad (105)$$

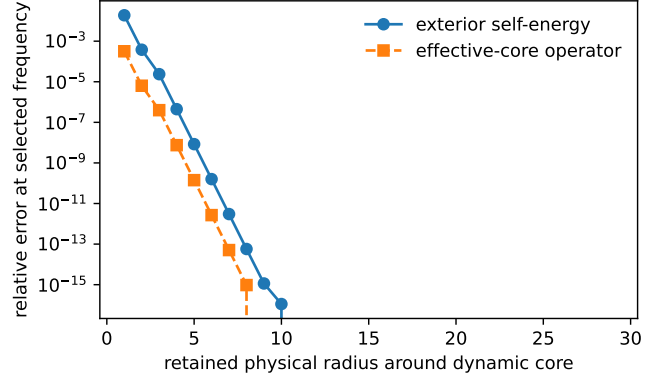


Fig. 4. Controlled sparse dynamic-network check. As the retained physical radius grows, the exterior self-energy and effective-core operator converge rapidly at the selected frequency. This illustrates the proposed boundary certificate; it is not a substitute for a detailed EMT benchmark.

This converts the vague question “how many neighboring buses belong in EMT?” into an operator-approximation problem at the ports of the dangerous core.

B. Minimal detailed region

For a frequency band Ω and weights selecting critical core directions, define

$$R^* = \arg \min_R |R| \quad \text{s.t.} \quad \sup_{\omega \in \Omega} \|W_L \Delta \Sigma_R(j\omega) W_R\|_2 \leq \varepsilon. \quad (106)$$

Structure-preserving model reduction and gap-metric ideas provide mature tools for constructing/passivating external equivalents rather than simply deleting the outside network [75]. Recent EMT–TS hybrid work independently identifies interface error and region expansion as a central accuracy issue [58]. Equation (106) supplies a mechanism-conditioned selection criterion: expand the detailed region only until the exterior seen by the dangerous core is accurate enough.

C. Three independent trust gates

Waveform quality alone cannot certify that a phasor model preserves fast inner-loop/filter dynamics. Recent work formulates phasor-device trustworthiness as a structured robust-stability problem and uses a frequency-resolved interaction index plus structured singular value μ to certify the effect of EMT-induced model mismatch [19]. The present workflow therefore uses three gates:

- 1) *waveform gate*: QSS/sinusoidality determine whether a phasor/CF description is kinematically meaningful;
- 2) *model-discrepancy gate*: structured uncertainty/ μ determines whether omitted EMT device dynamics can reverse the stability conclusion;
- 3) *mechanism-preservation gate*: (105)–(106) determine whether the chosen physical boundary preserves the self-energy/interaction core that matters.

The gates answer different questions and should not be collapsed into one scalar threshold.

D. Industrial output

The proposed planning product is now precise:

wide-area screen $\rightarrow$ connected action core
 $\rightarrow$ minimal self-energy-preserving EMT region
 $\rightarrow$ minimum recovery action

(107)

This chain is complementary to multiport dissipativity/root-cause methods, which already provide robust device-coupling diagnosis and mitigation [67], and to multiscale EMT algorithms that accelerate detailed simulation [77]. The distinctive target is finite portfolio non-composability together with a mathematically certified mapping from the action-space mechanism to the physical model boundary.

XVII. SPECTRAL DIFFERENTIAL GEOMETRY OF THE ACTION MANIFOLD

The equilibrium-manifold construction admits a compact geometric interpretation that clarifies what is path dependent and what is not. Fix a complex s and restrict attention to one connected region of action space on which $T(s, a)$ is nonsingular. Define the complex spectral one-form

$$\omega_s = \text{tr}(T^{-1} d_a T). \quad (108)$$

Locally,

$$\omega_s = d_a \log \det T, \quad (109)$$

so ω_s is exact and therefore closed: $d_a \omega_s = 0$. The finite determinant transport in Theorem 3 is simply the line integral of this exact one-form. Two smooth paths connecting the same actions give the same determinant ratio as long as neither crosses the spectral discriminant.

This observation has two practical consequences. First, the finite endpoint effect is not an artifact of the homotopy used to introduce a line or controller. A continuous susceptance ramp, for example, is a computational device for resolving where the interaction grows; the final determinant ratio is path independent within one nonsingular component. Second, topology becomes relevant when the action path encircles or crosses a singular set. The phase of $\det T$ can then acquire nontrivial winding even though the local one-form remains exact on each chart. The physically meaningful global integer is the argument-principle mode count, not a branch-dependent value of the complex logarithm.

A. Mode-count invariance and action homotopies

Let Γ be a closed contour in the s plane that is free of poles/zeros for all points on an action homotopy $a(\tau)$. Define

$$N_\Gamma(a) = \frac{1}{2\pi j} \oint_\Gamma \partial_s \log \det T(s, a) ds. \quad (110)$$

Then $N_\Gamma(a(\tau))$ is integer valued and continuous in τ , hence constant. A mode can enter or leave the protected region only when the homotopy intersects the discriminant (36). This gives a clean mathematical statement for a planning gate: if a reduced/interpolated model remains nonsingular on the contour throughout a certified model homotopy, the critical mode count is preserved.

B. Intrinsic versus feedback-mediated interaction

The second-order formula (41) can be split as

$$\kappa_{ij}^{\text{int}} = \text{tr}(GT_{ij}), \quad (111)$$

$$\kappa_{ij}^{\text{fb}} = -\text{tr}(GT_i GT_j). \quad (112)$$

The first term exists because the true re-equilibrated pencil is nonlinear in the action coordinates. It contains power-flow-mediated effects, parameter products, saturation of any smooth surrogate, and other direct mixed derivatives. The second term exists even for an affine pencil and is the genuine closed return through the system resolvent. A normalized mechanism share such as

$$\eta_{ij}^{\text{fb}} = \frac{|\kappa_{ij}^{\text{fb}}|}{|\kappa_{ij}^{\text{fb}}| + |\kappa_{ij}^{\text{int}}| + \epsilon} \quad (113)$$

is useful diagnostically, provided it is reported as a decomposition of the chosen spectral potential rather than a universal causal probability.

For third order, Theorem 4 separates a pure T_{ijk} vertex, three two-block terms, and two oriented three-return traces. This gives a vocabulary that is more informative than labeling all non-additivity “a cycle”: a finite externality may be dominated by equilibrium curvature, by a controller-mediated return, or by their interaction. The distinction can be tested numerically by automatic/implicit differentiation on the AC equilibrium manifold.

XVIII. FROM CONNECTED CUMULANTS TO SCALABLE PORTFOLIO COMPUTATION

The theory is useful only if it avoids exhaustive re-simulation of every action subset. The computational strategy exploits two structures: low-dimensional action support and connectedness.

A. Linear solves, not explicit inverses

No implementation should form the large matrix inverse $G = T^{-1}$. For local action blocks $B_i = U_i C_i V_i^*$, the port transports needed by the affine cycle formulas are

$$R_{ij}(s) = V_i^* X_j(s), \quad T_0(s) X_j(s) = U_j. \quad (114)$$

Thus one sparse/factorized solve per action input block constructs the entire low-dimensional return matrix. When many frequencies are required, sparse direct factorizations, rational Krylov spaces, or contour solvers can be reused. The same principle applies to $\text{tr}(GT_i)$ and higher contractions: solve against the low-rank factors and perform traces in port space.

B. Support-first screening

For m candidate actions, a naive exact coalition campaign requires 2^m system solves. The connected calculus first constructs a conservative interaction support graph from pair cumulants or port transports. After certified pruning, let the SCC sizes be $m_1, \dots, m_q$. Exact coalition evaluation then scales like

$$\sum_{c=1}^q 2^{m_c} \quad (115)$$

Algorithm 1 Connected finite-action screening

- 1: Solve the base AC equilibrium and construct the accepted sparse dynamic pencil $T(s, a_0)$.
- 2: Define physical action coordinates and compute total derivatives $D_i T$ using Proposition 2.
- 3: Build local port transports by sparse solves; estimate pair connected terms κ_{ij} over the frequency/security region.
- 4: Apply a distance/transport pruning rule only when the perturbation certificate (80) is satisfied.
- 5: Compute SCCs and, inside each SCC, simple-cycle overlap clusters and/or higher connected derivatives as required by (116).
- 6: For a candidate finite portfolio, evaluate the true re-equilibrated endpoint and audit the finite effect with Theorem 3.
- 7: For small dangerous cores, evaluate real coalition objectives and compute Möbius/Harsanyi dividends and Shapley allocation.
- 8: Pass the identified physical core to Algorithm 2 for detailed-region construction.

rather than 2^m , with additional savings when a cycle-cluster expansion is certified to low order. This is not an asymptotic guarantee for every grid: if the resolvent loses locality and one SCC contains all actions, the method correctly reports that the interaction problem is global.

C. Trace-order truncation with a rigorous gate

If $\|Q\|_2 = q < 1$, truncating (59) after order p yields

$$|R_p| \leq \frac{mq^{p+1}}{(p+1)(1-q)}. \quad (116)$$

The bound follows from $|\text{tr}(Q^k)| \leq m\|Q\|_2^k$. It is a convergence/tail certificate, not a stability boundary. If $q \geq 1$, the exact small-core determinant or a different expansion is evaluated instead of reinterpreting $q = 1$ as a security threshold.

*D. Algorithm 1: connected finite-action screening***XIX. A MODE-PRESERVING THEOREM FOR EMT BOUNDARY CERTIFICATION**

The self-energy formulation permits a stronger result than a heuristic norm tolerance: a reduced exterior can be certified to preserve the number of critical modes inside a contour.

Let the exact core operator after exterior elimination be $T_c(s)$ and the reduced/hybrid approximation be

$$\hat{T}_c(s) = T_c(s) + \Delta_c(s), \quad \Delta_c(s) = \Sigma_e(s) - \hat{\Sigma}_e^R(s). \quad (117)$$

Theorem 14 (Contour-preserving exterior reduction). *Let Γ be a closed contour on which $T_c(s)$ is nonsingular. If*

$$\sup_{s \in \Gamma} \|T_c^{-1}(s)\Delta_c(s)\|_2 < 1, \quad (118)$$

then T_c and $\hat{T}_c$ have the same net eigenvalue/zero count inside Γ under the adopted nonlinear-pencil bookkeeping.

Algorithm 2 Mechanism-preserving detailed-region selection

- 1: Start from devices/actions in the dangerous connected core plus their physical terminal buses.
- 2: Construct the exact or highest-fidelity available exterior self-energy on a frequency/contour grid.
- 3: Replace the exterior by the candidate RMS/phasor/reduced equivalent and evaluate Δ_c .
- 4: **if** Theorem 14 and the required I/O tolerance pass **then**
- 5: accept the current boundary;
- 6: **else**
- 7: add the physical buses/devices carrying the largest boundary self-energy error and repeat;
- 8: **end if**
- 9: Run detailed EMT on the accepted core and verify the predicted mechanism and recovery action.

Proof. Consider the homotopy $T_\tau = T_c + \tau\Delta_c$, $0 \leq \tau \leq 1$. On Γ ,

$$T_\tau = T_c [I + \tau T_c^{-1} \Delta_c]. \quad (119)$$

Condition (118) implies that the bracket is nonsingular for every τ by the Neumann/Banach lemma. Hence no zero crosses Γ along the homotopy. The argument-principle count is integer valued and constant, so the endpoint models have identical counts inside the contour. $\square$

Theorem 14 is directly aligned with the industrial question. A physical region is detailed enough when replacing its exterior cannot change the mode count in the protected damping/frequency domain. A structured- μ formulation can make the certificate less conservative when Δ_c has known block structure [19]; the theorem provides the unstructured baseline and the bridge to the exterior self-energy.

A. Frequency-band and trajectory objectives

A contour certificate protects modal topology, but EMT studies also care about peak current, voltage, current-limit activation, and protection. For an input/output map $H_c = C_c T_c^{-1} B_c$, the resolvent identity gives

$$\hat{H}_c - H_c = -C_c T_c^{-1} \Delta_c \hat{T}_c^{-1} B_c. \quad (120)$$

Hence self-energy error also bounds transfer-function error whenever both core operators are well conditioned. This gives a second region criterion for protection-relevant frequency bands. Time-domain waveform errors can then be evaluated only on the few candidate boundaries that pass the spectral screen.

XX. BLACK-BOX AND MULTI-VENDOR REALIZATION OF THE CALCULUS

The mathematical framework is written in terms of $T(s, a)$ and its action derivatives, not in terms of proprietary controller states. This permits a black-box realization provided the terminal dynamic models are identified with quantified uncertainty.

Recent work fits multivariable state-space models to black-box VSC frequency responses and recovers system-wide eigenvalue, sensitivity, and participation information in large

systems [62]. Gray-box identification similarly constructs interpretable reduced VSC models from terminal data [66]. These methods can supply the local dynamic blocks used by the present calculus without requiring firmware disclosure.

A. Measured action families

Suppose an OEM provides validated terminal models at operating/action samples $a^{(1)}, \dots, a^{(N)}$, for example several PLL bandwidths, GFL/GFM modes, or power levels. After network assembly and equilibrium reconciliation, the utility obtains

$$\hat{T}(s, a^{(k)}), \quad k = 1, \dots, N. \quad (121)$$

Total action derivatives can be estimated by local interpolation/finite differences on this *re-equilibrated model family*, rather than by differentiating hidden firmware. For example,

$$D_i T(s, a) \approx \frac{\hat{T}(s, a + h e_i) - \hat{T}(s, a - h e_i)}{2h}, \quad (122)$$

with higher mixed derivatives estimated only inside candidate connected cores. The finite endpoint/dividend calculations require only the identified coalition models and remain meaningful even when derivatives are too noisy for high order.

B. Uncertainty tubes instead of point models

A black-box realization should be carried as an uncertainty set

$$T(s, a) = \hat{T}(s, a) + B_\Delta(s, a) \Delta(s, a) C_\Delta(s, a), \quad \Delta \in \mathcal{D}. \quad (123)$$

The structured model-discrepancy gate can then use μ or dissipativity/IQC bounds to certify whether all admissible models preserve the relevant contour conclusion [19], [67]. Connected quantities are reported with uncertainty intervals or worst-case bounds, not as exact OEM-internal causal statements.

At the interaction level, a robust isolated-cycle reserve can be defined as

$$d_C^{\text{rob}}(s) = \inf_{\Delta \in \mathcal{D}} \min_{\nu \in \text{spec}((-1)^{|C|} \mathcal{H}_C(s, \Delta))} |1 - \nu|. \quad (124)$$

For an overlapping SCC, the corresponding robust object is the worst-case conditioning/zero count of $I + Q_S$. Exact computation may be expensive; structured singular-value bounds provide a conservative certificate. The industrial output is therefore privacy preserving in a precise sense: the utility needs validated terminal dynamics and uncertainty envelopes, while coordinate-invariant cycle spectra and finite project dividends can be computed without exposing local controller state labels.

C. Why this matters for interconnection studies

Multi-vendor studies frequently combine models supplied at different fidelity levels. The reconstructed framework separates three questions that are otherwise easy to mix: (i) whether the supplied terminal model is itself trustworthy for the frequency band, (ii) whether several project actions create a connected finite externality, and (iii) whether the surrounding network can be reduced without changing that mechanism. A project can

pass the first gate and still fail the second; a correct interaction diagnosis can pass the second and still be lost by an aggressive EMT/RMS boundary at the third. Treating these as independent certificates is a practical consequence of the mathematics, not an additional heuristic score.

XXI. POWER-SYSTEM PROBLEMS ADDRESSED BY THE RECONSTRUCTED THEORY

The theory is deliberately judged by decisions it can improve, not by the number of imported mathematical tools.

A. Interconnection portfolios and cluster externalities

Generator-interconnection studies increasingly evaluate many projects jointly. The relevant failure mode is not merely that a project has a bad standalone controller; it is that projects acceptable alone generate a new dynamic requirement only in combination. For a real security objective $F(S)$, the finite Möbius dividend

$$\mu(S) = \sum_{U \subseteq S} (-1)^{|S|-|U|} F(U) \quad (125)$$

is the irreducible project-set externality. The connected calculus provides a mechanism screen for which S can plausibly have large $\mu(S)$ and explains whether the effect is dominated by equilibrium curvature or closed controller feedback. Shapley allocation can then be used as an auditable engineering attribution of the chosen F , while any regulatory cost allocation remains a separate policy decision.

B. Topology planning under controller interaction

A transmission or collector reinforcement can improve a connectivity or short-circuit metric while moving a converter mode toward a controller resonance. The line itself is not “bad”; the joint action coordinate is. The re-equilibrated derivative (30) identifies whether the sign reversal is structural, operating-point mediated, or controller mediated. The finite dividend then tests whether two apparently beneficial upgrades are non-composable over the actual finite step.

C. Multi-vendor black-box grids

A utility may receive only terminal scans or validated reduced models from different OEMs. The action-quiver special case is useful precisely because its port blocks can be identified without exposing internal firmware. However, black-box compatibility is not enough: the model-fidelity gate must include an uncertainty envelope. The robust phasor-validity framework of [19] and multiport dissipativity of [67] are therefore complementary inputs. The connected calculus adds the portfolio question: which named OEM/project actions participate jointly in the externality and which finite retuning/connection choice breaks it?

TABLE III
ENGINEERING INTERPRETATION OF THE CORE MATHEMATICAL OBJECTS

Object	Question answered	Power-system interpretation
$D_i T$	What changes when action i is applied infinitesimally after re-equilibration?	Direct line/controller effect plus the global AC operating-point response.
κ_{ij}	Why do two actions fail to add?	Sum of intrinsic equilibrium curvature and a resolvent-mediated feedback return.
κ_S	Is there a genuine higher-order project interaction?	Connected multi-action density; lower-order disconnected effects are removed.
$\Delta_S F$	What finite effect exists only when all projects in S are present?	Exact finite dynamic externality for the chosen real planning objective.
Q / SCC	Where can affine/port feedback circulate?	Low-dimensional action-space core after certified localization.
$\mathcal{H}_C$	What does one traversal of a named MIMO feedback loop do?	Directional return map; its spectrum is independent of proprietary local state coordinates.
Σ_e	What dynamics does the omitted physical grid reflect back into the core?	Exact exterior dynamic equivalent at the core boundary.
r^-, r^+	How close are we to failure / how much change is needed to recover?	Weighted distance in physically implementable action coordinates.

D. Oscillation forensics

Participation factors answer where a mode appears in state coordinates; the wavemaker (93) answers where local feedback has leverage; endogeneity (92) partitions which modeled blocks contribute to the eigenvalue; connected action terms answer which finite engineering changes create the non-additivity. A post-event workflow should require consistency among all four layers before assigning responsibility.

E. Large electronic and computational loads

The 23-Hz Texas event is especially revealing because the instability was associated with a large electronic load and depended strongly on operating level [59]. This is the archetype of a re-equilibrated interaction problem: the relevant action coordinate may be load level, internal PFC/DC-link control, grid configuration, or neighboring converter tuning. A local “strong grid” label does not identify the feedback path. The proposed analysis would treat load/control changes as actions, construct the connected core around the observed frequency, and use the exterior self-energy to determine which surrounding grid dynamics must be preserved in EMT.

F. GFL/GFM and control-mode transitions

When a converter changes mode or control family, a purely infinitesimal derivative may be insufficient. The global endpoint identity (34) remains meaningful along a regular controller homotopy, while the finite real security objective is path independent at the endpoints. If a path approaches the spectral discriminant, the stability manifold itself constrains the feasible transition. This creates a natural interface between recent controller-stability-manifold work [69] and the connected mechanism calculus.

XXII. POWER-SYSTEM INTERPRETATION OF THE NEW MATHEMATICAL OBJECTS

For practitioners, the objects introduced above can be read without the full combinatorial machinery.

The central distinction is between a *property* and a *relation*. SCR, a CF-strength indicator, a local damping coefficient, or a participation factor attaches information to a bus/device/mode. A connected cumulant or finite dividend attaches information to a *set of interventions*. This is why the framework is not naturally reducible to one more nodal index.

XXIII. WHY LOW INERTIA CHANGES THE INTERACTION PROBLEM

The scaling arguments in this section concern the affine tangent-cycle terms of Corollary 5. They are local mechanism laws, not a claim that the full re-equilibrated finite intervention obeys the same power law without integration along the operating manifold.

An initially attractive hypothesis was that lower inertia always makes action portfolios more non-composable. The experiments falsified that statement. The direction depends on what happens to damping at the same time.

Consider a dominant isolated network mode with modal mass m_k , damping d_k , and stiffness λ_k . Under uniform co-retirement

$$M_\alpha = \alpha M_0, \quad D_\alpha = \alpha D_0, \quad (126)$$

its natural frequency scales as $\omega_k \propto \alpha^{-1/2}$. At $s = j\omega_k$, the inertial and stiffness terms cancel in the modal dynamic stiffness and

$$|t_k(j\omega_k)| \propto \alpha^{1/2}, \quad |G_k(j\omega_k)| \propto \alpha^{-1/2}. \quad (127)$$

A closed action cycle of order p containing p Green factors therefore obeys

$$|K^{(p)}| \propto \alpha^{-p/2}. \quad (128)$$

Pair and triple cycles scale as α^{-1} and $\alpha^{-3/2}$.

If inertia is reduced while *absolute damping is preserved*, $D_\alpha = D_0$, the damping term scales oppositely at the resonance frequency, yielding

$$|G_k| \propto \alpha^{1/2}, \quad |K^{(p)}| \propto \alpha^{p/2}. \quad (129)$$

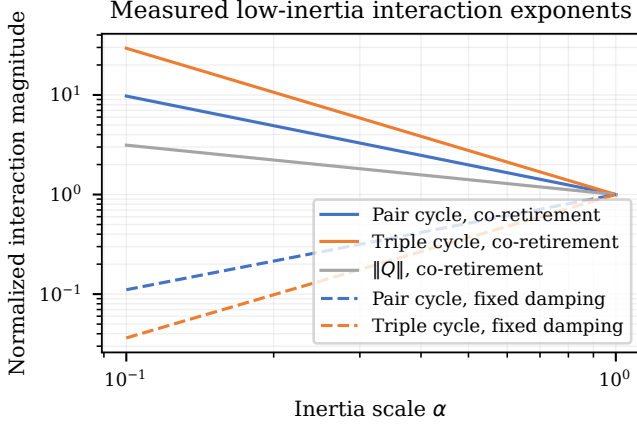


Fig. 5. Normalized fitted laws from the low-inertia campaign. The result is a joint inertia–damping law, not an “inertia alone” law. Co-retirement amplifies higher-order cycles fastest; preserving absolute damping reverses the trend.

Thus the composability trend reverses. In 160 random low-damping isolated-mode networks, measured slopes under co-retirement were -0.989 (pair), -1.469 (triple), and -0.496 for $\|Q\|$, with $R^2 > 0.99989$. With fixed damping, the slopes were $+0.956$, $+1.440$, and $+0.488$.

This scaling has a direct planning interpretation. The energy transition does not automatically make all interactions worse because inertia disappears; interactions intensify when dissipative support is retired with it. Conversely, procuring or preserving absolute damping can reduce interaction complexity even as inertia declines. This is a more precise industrial statement than a generic “low inertia is fragile” claim.

XXIV. WORKED THREE-ACTION EXAMPLE: A PURE HIGHER-ORDER EXTERNALITY

A small example makes the portfolio claim transparent. Consider three already self-normalized scalar actions with

$$L = \begin{bmatrix} 0 & 0 & -0.88 \\ 0.92 & 0 & 0 \\ 0 & 0.95 & 0 \end{bmatrix}. \quad (130)$$

The only closed cross-action path is

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 1, \quad h_C = (-0.88)(0.92)(0.95) = -0.7691. \quad (131)$$

Every one-action and every two-action principal submatrix is acyclic. Therefore

$$\det(I + L_{\{i\}}) = 1, \quad \det(I + L_{\{i,j\}}) = 1 \quad (132)$$

for all i, j , and an isolated or pairwise interaction screen reports no cross-action correction. The three-action portfolio, however, gives

$$\det(I + L) = 1 + h_C = 0.2309. \quad (133)$$

Thus the entire interaction is a third-order dividend supported on one prime cycle. No assignment of independent scalar “project risk scores” can reconstruct this fact unless the three-way configuration is evaluated or the cycle structure is retained.

A. Physical interpretation in an interconnection cluster

The three vertices can be read as: (1) a new collector/POI topology, (2) a GFL PLL/current-controller setting, and (3) a BESS/GFM plant-control mode. The negative sign in (131) represents the net signed phase/gain closure on the declared spectral contour or frequency point; it need not come from a negative physical susceptance. All three projects can have acceptable isolated margins because the destructive return path is incomplete until the third action closes it.

This example clarifies why the proposed framework is not merely a higher-order sensitivity expansion. The value 0.2309 follows from a finite determinant and remains valid even when the action magnitudes are not infinitesimal, provided the accepted base/action representation remains valid. A first-order eigenvalue derivative can warn that each individual parameter matters, but it does not by itself identify that the first nonzero *interaction* lives on a three-action closed path.

B. Cycle surgery is immediately interpretable

Suppose only the third edge can be modified and the engineering objective is to restore

$$\det(I + L) \geq \delta, \quad \delta = 0.50. \quad (134)$$

Writing $L_{13} = -x$ gives $1 - 0.92 \times 0.95 x \geq 0.50$, so

$$x \leq \frac{0.50}{0.92 \times 0.95} = 0.5721. \quad (135)$$

The action-space result therefore says exactly which transport must be weakened and by how much in this stylized example. In a real grid, x would be a nonlinear function of PLL bandwidth, droop, virtual damping, line/POI topology, and operating point. The cycle derivative provides the direction; the constrained physical optimization in Section XV maps that transport change back to feasible engineering variables and then hands the candidate to full EMT.

C. The MIMO version adds directional memory

If each action port is two-dimensional, the scalar h_C is replaced by $\mathcal{H}_C \in \mathbb{C}^{2 \times 2}$. Two portfolios can have similar spectral radius $\rho(\mathcal{H}_C)$ but very different eigenvectors and commutators with neighboring cycles. This directional information is precisely what disappears in scalar grid-strength scores. A controller retuning may therefore leave the size of one loop nearly unchanged while rotating its return direction so that it no longer reinforces another loop. Non-Abelian holonomy makes that mode-mixing mechanism visible.

XXV. WHERE THE PROPOSED CALCULUS FITS RELATIVE TO EXISTING POWER-SYSTEM TOOLS

The novelty boundary becomes clearer when the framework is placed next to the tools an interconnection engineer already has. The proposed method should not claim to replace any mature method that directly answers a better-defined question.

TABLE IV
FUNCTIONAL GAP RELATIVE TO COMMON POWER-SYSTEM ANALYSIS LAYERS. “MISSING” MEANS NOT NORMALLY PROVIDED AS THE PRIMARY OUTPUT OF THAT LAYER, NOT THAT NO VARIANT IN THE LITERATURE COULD BE ENGINEERED TO PROVIDE IT.

Layer	Excellent at	Typical decision object	Gap relevant here	Role in proposed workflow
SCR/ESCR/graph strength	Fast structural screening and weak-grid intuition	Bus/POI strength	Controller-specific finite portfolio externality; ordered MIMO feedback	Upstream screen; never discarded, but gated by controller relevance.
Eigenvalues, participation, sensitivities	Identifying critical modes/states and local parameter derivatives	Mode, state, parameter	Finite multi-action composability and exact higher-order interaction support	Base modal evidence and gradients; action cycles explain joint finite changes.
Impedance/multiport/Nyquist	Frequency-domain converter interaction and return-ratio stability	Device/port loop	Cluster action attribution and finite corrective-action game	Supplies/validates port blocks; quiver organizes multiple engineering interventions.
Robust μ /uncertainty analysis	Worst-case stability under structured uncertainty	Uncertainty block	Physical responsibility and named cycle mechanism are secondary	Natural robustification layer for uncertain OEM ports.
RMS/positive sequence	Wide-area nonlinear dynamics at tractable cost	Contingency/trajectory	Fast controls/protection and some converter interactions may be omitted	Wide-area base and first-stage screen.
Detailed EMT	Highest-fidelity controller/protection interaction available to planner	Scenario waveform/trajectory	Expensive model assembly, study-area selection, limited automatic mechanism attribution	Final authority on escalated dynamic core and mitigation.
Action-quiver calculus	Finite action composability, feedback-core localization, cycle attribution	Project/controller action SCC and cycle	Requires a valid underlying model; not a waveform-certification tool	Orchestrates when/where EMT is spent and what correction is tested first.

A. Relation to impedance and robust multi-vendor stability

Multiport impedance and generalized Nyquist methods already provide rigorous ways to expose converter interactions, while robust-stability work explicitly addresses uncertain multi-vendor controls [43]. The action-quiver contribution is orthogonal: its vertices are not required to be individual physical converters. A vertex can be a proposed line, an alternate POI, a controller-mode change, or a whole project. The determinant restriction property then asks a planning question that impedance analysis alone does not automatically answer: *which finite set of engineering decisions must coexist before the harmful closed path exists?*

Conversely, the quiver should use impedance/EMT information whenever it is the better port model. Nothing in the determinant construction requires an electromechanical state-space realization. An identified or frequency-scanned MIMO return map can supply the edge transports, provided the action restriction and base-model assumptions are documented. This makes the framework a possible organizing layer over heterogeneous vendor models rather than a competitor to them.

B. Relation to eigenvalue sensitivity and participation

Participation factors answer which states or devices participate in a mode; eigenvalue sensitivities answer how an infinitesimal parameter change moves that mode. The action calculus asks a different counterfactual: if several declared finite modifications are applied, does the characteristic correction factorize into isolated effects or does a closed cross-action feedback core appear? The two views are complementary. In fact, the full nonlinear-eigenvalue sensitivity of Section IV can rank candidate action variables within a cycle after the cycle is located.

C. Why the gap is especially acute in cluster interconnection

A serial study can often focus on one project’s incremental impact against a fixed prior network. Cluster studies intentionally evaluate many projects and network upgrades together. As the number of power-electronic projects grows, the number of possible controller/topology combinations grows much faster than the number of projects. The SCC and cycle representation is valuable precisely because it seeks structural zeros in that combinatorial space: projects outside the same directed feedback core have zero determinant interaction under the model, while only the small closed core requires combinatorial attention.

XXVI. EVIDENCE: FROM ALGEBRAIC CHECKS TO IEEE 39-BUS DECISIONS

The project contains several evidence tiers. Keeping them separated is essential for claim calibration.

A. Exactness and numerical identity checks

The Schur/full-state determinant identity was evaluated at 100 random complex points in the controller mechanism model, with maximum relative discrepancy 5.8×10^{-15} . Acyclic and SCC determinant factorizations were tested on 500 random instances with numerical errors no larger than approximately 2.22×10^{-16} . The new re-equilibrated transport identity was separately audited on a nonlinear three-action toy problem: the path integral matched the finite log-determinant change to 5.7×10^{-16} and the finite three-action Möbius dividend matched the integral of the mixed third derivative to 1.35×10^{-10} . These checks validate the implemented identities, not the fidelity of a converter model.

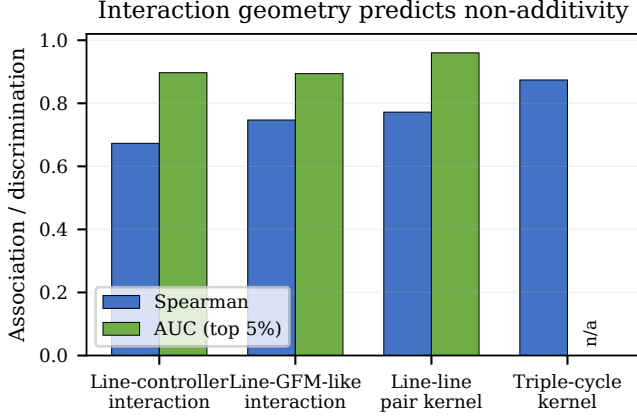


Fig. 6. Empirical bridge from exact spectral interaction geometry to damping-margin non-additivity. Pair/triple kernels are not perfect surrogates for $\zeta_{\min}$, but they provide useful screening fidelity.

B. Mixed line–controller non-additivity

For a line reinforcement e and controller retuning c , define finite damping-margin non-additivity

$$\mu_{ec} = \zeta_{ec} - \zeta_e - \zeta_c + \zeta_0. \quad (136)$$

An exact determinant interaction observable was also computed:

$$I_{ec}(s) = \log |\det T_{ec}(s)| + \log |\det T_0(s)| - \log |\det T_e(s)| - \log |\det T_c(s)|. \quad (137)$$

Across 17,664 line/controller pairs, $|I_{ec}|$ had Spearman correlation 0.673 with $|\mu_{ec}|$ and AUC 0.897 for identifying the strongest 5% non-additivities. A separate 23,000-pair line–GFM-like campaign produced Spearman 0.747 and AUC 0.894. Hybridization was strongly associated with the maximum mixed-action interaction across operating points (Spearman 0.920), supporting the causal chain

$$\begin{aligned} \text{hybridization} &\rightarrow \text{large action coupling} \\ &\rightarrow \text{non-composable portfolio}. \end{aligned} \quad (138)$$

The flagship destructive portfolio is deliberately easy to interpret. The baseline minimum damping ratio is

$$\zeta_0 = 0.045218. \quad (139)$$

Reinforcing line 26–29 yields $\zeta_e = 0.045872$ (+1.45%), and retuning the controller natural frequency from 5.1 to 4.8 rad/s yields $\zeta_c = 0.047657$ (+5.39%). Applying both actions gives

$$\boxed{\zeta_{ec} = 0.040352 \quad (-10.76\% \text{ relative to baseline}).} \quad (140)$$

The finite interaction is $\mu_{ec} = -7.96 \times 10^{-3}$. The hybridization indicator rises from 0.381 at baseline to 0.428 with the line, 0.982 with the retune, and 1.121 jointly. Figure 7 shows the planning error that an isolated-action workflow would make.

The qualitative phenomenon was not confined to one tuned pair. With moderate actions, 11.5% of 200 randomized high-IBR scenarios contained at least one destructive line–GFM-like pair; under a stressed campaign the fraction was 23.5%. These percentages are not presented as field occurrence rates. Their role is only to demonstrate that the mechanism survives randomization.

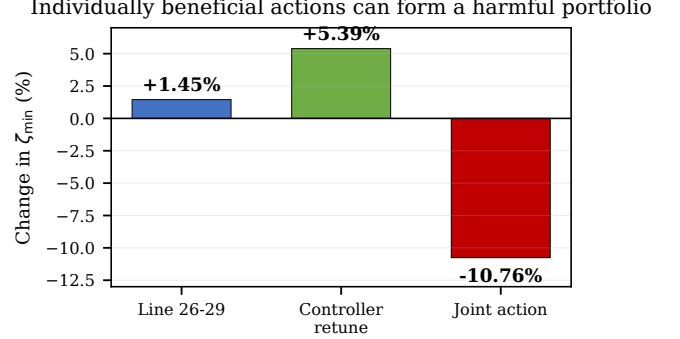


Fig. 7. Flagship controller-mediated portfolio counterexample. Each action is beneficial when screened alone; their joint implementation is strongly detrimental.

C. Topology-only pair and higher-order interference

The action-cycle picture also appears without an explicit controller retune. In one high-IBR case, line 19–20 alone improved $\zeta_{\min}$ by 15.8% and line 20–34 alone by 9.1%, yet their joint reinforcement reduced the margin 7.55% below baseline. The three highest-ranked individual line actions together reduced the margin by 6.87%, whereas an interaction-aware triple improved it by 19.59%.

For $F(S) = \zeta_{\min}(S)$, both positive and negative Möbius interactions were observed, with violations of monotonicity. Consequently, this damping objective is not universally monotone, submodular, or supermodular in the tested regime. Greedy guarantees from unrelated actuator-selection problems cannot be imported without verifying the appropriate set-function structure.

The Green pair kernel

$$\mathcal{K}_{ef} = |\beta_e \beta_f r_{ef} r_{fe}| \quad (141)$$

had Spearman 0.772 with $|\mu_{ef}|$ and AUC about 0.96 for detecting the strongest 5% pair interactions. For 286 triples, the corresponding three-cycle kernel had Spearman 0.874 with the genuine third-order Möbius interaction. These statistics provide an empirical bridge between local/affine cycle kernels and a planning objective such as damping margin. The exact finite externality for the re-equilibrated problem is the coalition Möbius difference or, equivalently, the integral identity in Theorem 6.

D. Exact responsibility on the destructive pair

For direct damping-margin attribution inside a two-action core, the four coalition evaluations are affordable:

$$F_0 = 0.045218, \quad F_e = 0.045872, \quad (142)$$

$$F_c = 0.047657, \quad F_{ec} = 0.040352. \quad (143)$$

The margin-game Harsanyi terms are

$$\mu_e = 0.000654, \quad \mu_c = 0.002439, \quad \mu_{ec} = -0.007959. \quad (144)$$

TABLE V
IEEE 39-BUS + THREE-GFL CLUSTER: TOPOLOGY COMPARISON AT 95 MW PER PROJECT

Topology	$\zeta_{\min}$ (%)	Mode (Hz)	Fiedler	$\rho(H_+)$	Max load (%)	$G_{\max}$
Closed triangle, 1 POI	2.876	7.51	5.117	1.388	97.88	8.09
Radial, 1 POI	5.152	7.98	3.556	0	97.63	8.09
Closed triangle, 2 POIs (25,29)	4.004	8.10	7.542	1.683	93.97	7.88
Radial, 2 POIs (25,29)	5.327	2.64	7.405	0.0177	92.26	7.95
Closed triangle, 3 POIs	2.132	8.84	9.130	0.958	89.44	7.83
Radial, 2 POIs (25,26)	5.326	2.64	8.549	0.0385	89.34	8.06

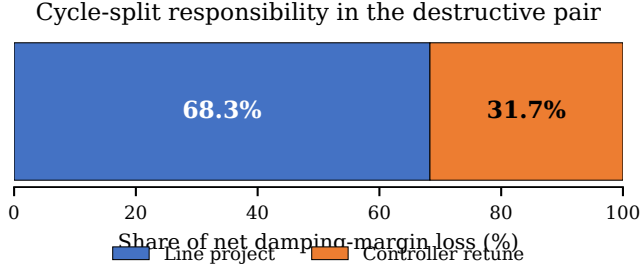


Fig. 8. Direct two-action Shapley allocation of the net damping-margin loss in the flagship destructive core. The cycle formula is exact for the determinant game; for $\zeta_{\min}$, exact coalition evaluations are used here.

The two-player Shapley values become

$$\phi_e = \mu_e + \frac{1}{2}\mu_{ec} = -0.003325, \quad (145)$$

$$\phi_c = \mu_c + \frac{1}{2}\mu_{ec} = -0.001541. \quad (146)$$

Thus the net loss is allocated 68.3% to the line project and 31.7% to the controller retune. Figure 8 visualizes the split.

This distinction is important. For a real engineering objective such as $\zeta_{\min}$, Shapley values are computed from real coalition evaluations (or from an explicitly validated surrogate). The determinant interaction is a mechanistic spectral observable, while Theorem 6 supplies the exact mathematical bridge between finite Möbius effects and connected mixed derivatives when the chosen scalar objective is smooth.

E. IEEE 39-bus + three-GFL connection study

The strongest topology-planning example attaches a 285-MW three-project GFL cluster to MATPOWER case39. Buses 40–42 inject 95 MW each at 345 kV. For each topology, the nonlinear AC power flow is re-solved; the full P – Q injection Jacobian is formed; constant- P/Q load buses are Schur-eliminated; and angle, frequency, voltage, PLL, and reactive-control states are retained at dynamic ports. The GFL blocks are transparent reduced surrogates with nominal $K_p = 13$, $K_i = 400$, not manufacturer models.

Table V contains the principal configurations. Every case shown is AC feasible and asymptotically stable, but damping is not monotone in scalar connectivity.

The closed single-POI triangle has 43.9% larger retained-port Fiedler connectivity than the radial single-POI design, but minimum damping falls from 5.152% to 2.876%, a 44.2% degradation. Maximum branch loading changes by only 0.25

percentage points. The three-POI closed loop has the *largest* Fiedler value in the table and the *worst* damping. Figure 10 makes the contradiction visible.

To isolate causality, the 42–40 closing branch was swept continuously from open to fully closed. The conventional 5% damping screen was crossed at a closing fraction near 0.543, although the system remained asymptotically stable. The reduced project-cycle return magnitude became order-one earlier, around 0.488. This ordering makes it a useful empirical precursor in that experiment, but $\rho(H_+) = 1$ is *not* asserted as a universal stability boundary.

The effect also occupies a finite controller/network region: a 31×33 sweep of loop closure and PLL K_p produced 1023 AC-relinearized cases. For every sampled K_p between 12.5 and 24, the radial configuration met the 5% screen while the fully closed loop did not. A separate closure/POI-strength campaign showed that radialization alone cannot rescue a genuinely weak external connection. Two mechanisms must be separated: internal controller-cycle closure and external grid strength.

F. Mode localization and physical transients

Biorthogonal participation localizes 99.52% of the least-damped closed-loop mode to buses 40–42, with 97.71% on buses 41–42. The relevant poles are

$$\lambda_{\text{loop}} = -1.358 \pm j47.195 \text{ s}^{-1}, \quad \zeta = 2.876\%, \quad (147)$$

$$\lambda_{\text{radial}} = -2.588 \pm j50.163 \text{ s}^{-1}, \quad \zeta = 5.152\%. \quad (148)$$

The corresponding amplitude decay constants are approximately 0.736 s and 0.386 s. Closing one physical collector line nearly doubles the decay time of the project-local GFL mode.

A 10-MW, 100-ms active-power pulse at bus 40 gave a useful negative result. The closed-loop case had maximum project frequency deviation 0.152 Hz, maximum PLL angle error 1.93° , and maximum voltage deviation 1.44×10^{-3} p.u.; the recommended radial 25–26 case had 0.175 Hz, 2.36° , and 7.21×10^{-4} p.u. The forced pulse did not discriminate the architectures as strongly as modal damping. Similarly, the physical maximum transient energy gain stayed near eight across candidate topologies even as damping ranged from 2.13% to 5.33%. Therefore the observed architecture reversal is primarily a modal/controller-cycle phenomenon, not a demonstrated non-normal Braess effect.

G. Connection search

With bus 25 fixed as the first POI, radial second POIs were tested at buses 3, 17, 26, 28, 29, and 39. Several passed the

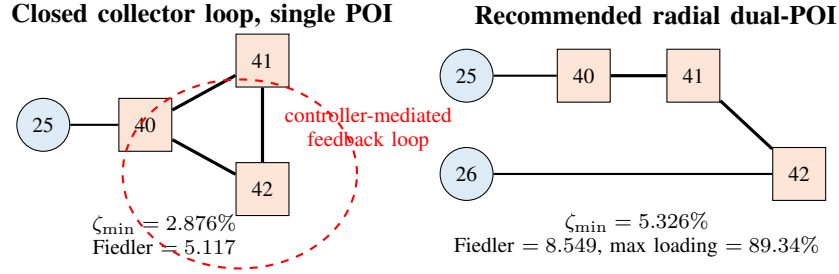


Fig. 9. IEEE-39+three-GFL mechanism experiment. The internal closing edge creates a controller-mediated project loop. Removing that edge and using a second external point of interconnection (POI) restores the damping screen while improving branch loading.

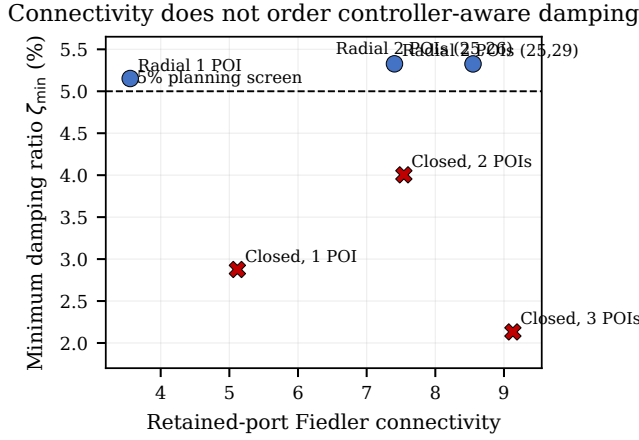


Fig. 10. The IEEE-39+three-GFL study falsifies a monotone scalar-connectivity–damping ordering for the modeled controller-aware system. Radial architectures lie above the 5% screen even when some have lower Fiedler values than closed-loop alternatives.

damping screen. The local 25–26 design provides a balanced recommendation: $\zeta_{\min} = 5.326\%$, Fiedler = 8.549, maximum branch loading = 89.34%, and small residual triangle-mediated return magnitude 0.0385. Buses 17 or 3 reduced loading further but imply more remote infrastructure. The scientific contribution is not simply “radial is better”: the project identifies the specific controller-mediated loop, shows that connectivity improvement and damping deterioration coexist over a finite region, and provides a topology change that removes the mechanism without removing the generation projects.

XXVII. VALIDATION ARCHITECTURE AND FALSIFICATION PROGRAM

A theory paper must distinguish exact identities from engineering validation. The validation ladder is therefore organized around falsifiable questions.

A. Gate A: re-equilibrated identities

For every finite action campaign, recompute the nonlinear AC equilibrium and audit Theorem 3 by numerical path integration. Compare finite Möbius differences against the integral of mixed connected derivatives in Theorem 6. A frozen low-rank approximation is reported separately so its breakdown is visible rather than hidden.

B. Gate B: closest strong baselines

The flagship “strong grid–weak interaction core” experiment must compare against SCR/gSCR-type quantities, CF strength $S^{(0:2)}$, full modal/impedance information, dissipativity or another robust multiport certificate, and ordinary eigenvalue sensitivity. Fiedler connectivity remains a useful visualization but is no longer treated as the main adversary.

C. Gate C: scalability and locality

For 10–100+ candidate actions, report raw action-graph density, certified pruned density, SCC size distribution, number/size of connected cycle clusters, resolvent solves, runtime, and error in the finite interaction observable. If the certified graph remains one giant SCC, the claimed combinatorial scaling benefit must be withdrawn for that regime.

D. Gate D: physical boundary selection

A detailed EMT or validated average-value model provides the reference. Compare at least four region-selection rules: fixed k -hop, electrical distance, strength-based selection, and the self-energy-preserving criterion (106). Report error in critical mode/frequency, peak current/voltage, protection-relevant quantities, and runtime versus retained model size. The mechanism-preserving method succeeds only if it attains a better accuracy–cost Pareto frontier, not merely if it identifies a plausible region.

E. Gate E: cycle surgery

For every strong mechanism claim, modify an action on the predicted connected core and a matched action outside it. The method should predict the sign and approximate magnitude of the modal/security change. A negative control outside the core is necessary before using strong mechanistic language.

F. Modern external benchmark

The real 23-Hz large-electronic-load event provides an unusually relevant external target: oscillations emerged near 320 MW and detailed EMT reproduced the recorded behavior [59]. A future implementation should test whether the connected-action/self-energy machinery localizes the grid–load feedback and whether the post-event controller/firmware change can be represented as a recovery action. The present manuscript cites this event as a target, not as an internally reproduced result.

XXVIII. FALSIFICATION-FIRST VALIDATION PROGRAM FOR A TRANSACTIONS SUBMISSION

The mathematical reconstruction is only useful if its strongest claims can fail cleanly. The validation program is therefore organized around comparisons that would disprove, rather than merely illustrate, the proposed mechanism.

A. Gate A: *re-equilibrated calculus versus frozen low-rank surrogates*

For each finite line/controller/POI action, the nonlinear AC equilibrium is re-solved. Three predictions are then compared: 1) the exact endpoint change from the re-linearized full model; 2) the path integral in Theorem 3; 3) the frozen affine/quiver approximation and its adequacy bound (52).

The theory fails this gate if the exact path integral does not reproduce the endpoint determinant change numerically, or if the proposed curvature bound fails to cover the frozen-model error. A useful result is not that the affine approximation is always bad; it is that the gate predicts when it is safe to use.

B. Gate B: *mechanism identification against strong baselines*

The principal comparison should not use Fiedler connectivity alone. For the same operating points and actions, the study should report conventional SCR/gSCR where defined, the zero/first/second-order complex-frequency strength indicators, modal/impedance participation, a multiport dissipativity or return-ratio diagnosis, and the connected action quantities. The claim is supported only if the connected calculus adds information unavailable from the competing diagnosis: finite portfolio non-additivity, a higher-order action set, or a distinct minimum recovery action. Existing impedance/dissipativity methods must be credited whenever they identify the same root cause.

C. Gate C: *cycle surgery and negative controls*

A candidate mechanism C^* or connected SCC is accepted only after intervention surgery. The experiment perturbs one member of the predicted core, recomputes the full model, and checks the predicted change in damping/contour reserve. A matched perturbation of comparable engineering magnitude outside the identified core acts as a negative control. Repeating this across operating points distinguishes a mechanism from a post-hoc participation map.

D. Gate D: *scalability and certified locality*

For increasing numbers of candidate actions, report the unpruned action density, decay of port transports with physical distance, nonnormality/conditioning factors, certified pruning error, SCC-size distribution, number of connected cycle clusters retained, sparse linear-solve count, and total runtime. If the resolvent becomes global, the method must report a large core rather than forcing a favorable complexity result. A strong scaling figure should compare exhaustive 2^m coalition evaluation with the connected/SCC strategy on cases where exact enumeration is still possible.

E. Gate E: *model boundary versus full electromagnetic reference*

The industrial headline requires an electromagnetic reference. The full detailed model is treated as ground truth for the tested model family. Candidate detailed regions are selected by (i) k -hop distance, (ii) electrical distance, (iii) strength-based heuristics, and (iv) the exterior-self-energy certificate. Each boundary is evaluated on critical mode count, oscillation frequency/damping, terminal transfer functions, peak current/voltage, limit activation, and runtime. The proposed method is useful only if it achieves a better fidelity–cost Pareto frontier, not merely a smaller region.

F. Gate F: *a second modern-grid benchmark*

The IEEE 39-bus mechanism is necessary but insufficient. A second benchmark should involve a large electronic/computational load or a multi-vendor converter cluster. The recent 23-Hz large-electronic-load case provides a particularly relevant target [59]: the method should test whether the observed frequency band is associated with a compact connected action core, whether local strength remains a poor proxy for that mechanism, and whether the self-energy boundary predicts which surrounding dynamics must be retained. Failure to recover a meaningful core would narrow the claimed industrial scope.

This validation architecture is intentionally demanding. The present reduced IEEE 39-bus studies provide mechanism evidence, not completion of Gates E–F. The reconstructed theory is designed so that the missing electromagnetic evidence is a clear external experiment, not a hidden assumption inside the mathematics.

XXIX. DISCUSSION

A. *Why the action-space view is the main conceptual shift*

Most traditional grid metrics assign a property to a node, line, or mode. The intervention formulation instead asks about a *relation among candidate decisions*. This is not a semantic change. A line reinforcement can have a positive isolated margin, a controller retune can have a positive isolated margin, and yet the pair can close a high-gain feedback cycle. No scalar score attached separately to the line and controller can encode that all-order relation unless it is recomputed jointly.

The action-space determinant also separates two reasons a cheap planning workflow can fail. First, the base model may be too crude: omitted converter states change G_0 . That is the self-energy/fidelity problem. Second, even a correct G_0 may be used incorrectly if actions are ranked independently despite strong off-diagonal K_A . That is the composability problem. Figure 1 separates the model-fidelity and action-composability levels because fixing one does not fix the other.

B. *Relation to Braess-type reasoning*

The project repeatedly encountered cases in which adding a line increased a graph-connectivity quantity but worsened damping. It is tempting to label all such results “Dynamic Braess.” That label is too broad. Damping ratio can decrease

TABLE VI
DECISIVE VALIDATION MATRIX: CLAIM, BASELINE, AND FALSIFICATION CONDITION

Claim	Required comparison	Pass evidence	Falsification / scope reduction
Re-equilibrated finite calculus	Re-solved endpoint vs. path integral vs. frozen action	Endpoint/path agreement; quiver error inside certified bound	Path identity fails numerically or frozen-error certificate is violated
Higher-order portfolio externality	Single actions, all pairs, full coalition	Nonzero finite dividend tied to connected derivative/core	Joint effect explained entirely by lower-order marginals
Mechanistic action core	Participation, impedance/return-ratio, dissipativity	Cycle surgery succeeds and negative control is weak	Outside-core perturbation is equally predictive or mechanism is not reproducible
Certified locality	Exact dense action operator	Pruned determinant/core quantities remain inside error bound	Required tolerance forces one giant SCC
Minimal detailed region	Full electromagnetic model and heuristic boundaries	Same protected mode count plus better fidelity/cost Pareto point	Heuristic region is smaller/equally accurate or certificate misses a critical mechanism
Minimum recovery	Brute-force/optimization candidates	Predicted r^+ action restores margin in full detailed model	Full model does not recover or a cheaper feasible action exists

simply because a frequency rises faster than decay rate; a controller-mediated line can also change operating point, rotate modes, or activate a feedback loop. A defensible Braess-style statement requires a clearly defined performance metric, a network modification that appears beneficial under an appropriate structural notion, and a nontrivial interaction mechanism explaining the degradation [10]. The 39+3 loop case satisfies a stronger version of this narrative because the internal controller cycle is identified and removable, but the paper should still lead with controller-aware cycle closure rather than the Braess label.

C. Relation to non-normality

Non-normality remains an important independent vulnerability in converter-rich grids [9], [20]. Yet the strongest topology example here barely changes physical maximum transient energy gain while strongly changing modal damping. This separation is scientifically useful: it prevents the project from forcing all phenomena into one “hidden transient” story. Modal/cycle degradation and non-normal amplification can coexist and should be treated as separate security layers.

D. What makes the attribution result unusually valuable

The prime-cycle Shapley result is potentially the most distinctive mathematical bridge in the project because it converts a generic computational problem into a structured one. General Shapley evaluation is expensive because every coalition can carry an independent value. Here the characteristic function derives from a perturbation determinant with principal-submatrix restriction. Pure synergies therefore live on feedback cycles and can be enumerated locally in SCCs. The power system supplies the Green operator; graph-cycle algebra supplies the decomposition; cooperative-game theory supplies the fair-allocation axioms. The combination is more than a juxtaposition because the determinant identifies the game dividends exactly.

E. Limits of causal language

The word “causal” should be used in an engineering, intervention-defined sense: execute a declared set of finite actions against a common base, and attribute the resulting spectral interaction to closed action cycles. The framework does

not infer causal structure from observational data, nor does it identify market/legal liability. It provides counterfactual but-for structure within a specified physical model. That boundary should remain explicit in any journal or regulatory presentation.

XXX. CONCLUSION

This paper replaces a frozen low-rank intervention picture by a connected calculus on the re-equilibrated operating manifold. The key result is structural: finite power-system actions can be studied through mixed connected derivatives of a physical pullback pencil, and in the affine/low-rank limit those derivatives reduce exactly to controller-mediated action cycles. Finite Harsanyi/Möbius externalities are integrals of the same connected densities, so differential interaction, finite portfolio non-additivity, and real-valued attribution belong to one theory rather than to separate patches.

The reconstruction also clarifies the role of cycle language. Determinants involve finite disjoint cycle systems; logarithms involve closed walks and connected overlapping-cycle clusters; MIMO cycle holonomies are meaningful through similarity invariants and the algebra they generate. Certified locality is required before SCC compression is trusted. Complex-frequency strength remains a local temporal diagnostic, not a surrogate for global interaction closure.

For power systems, the most important consequence is the model-boundary formulation. Once a dangerous connected core is identified, the exterior network should be reduced only insofar as its Schur self-energy at the core boundary is preserved. This converts EMT study-area selection from a geographic heuristic into a mechanism-preservation problem that can be combined with structured model-mismatch certificates. The IEEE 39-bus studies demonstrate the motivating non-additivity and connection-design mechanism at reduced-model fidelity; decisive industrial claims require the validation gates stated explicitly in Section XXVII.

The intended contribution is therefore narrow but potentially consequential: determine when finite engineering interventions fail to compose, expose the connected controller–network mechanism responsible, preserve that mechanism in the smallest credible detailed model, and compute the minimum physically realizable path back to a required security margin.

APPENDIX A HIGHER PULLBACK DERIVATIVES ON THE AC EQUILIBRIUM MANIFOLD

The first-order pullback derivative in Proposition 2 already shows why a frozen structural update is not the correct finite object. At second order the distinction becomes explicit. Write $\chi_i = \partial\chi/\partial a_i$ and $\chi_{ij} = \partial^2\chi/(\partial a_i\partial a_j)$. Differentiating

$$H_y\chi_i + H_{a_i} = 0 \quad (149)$$

with respect to a_j gives

$$H_y\chi_{ij} + H_{yy}[\chi_j, \chi_i] + H_{ya_j}\chi_i + H_{a_iy}\chi_j + H_{a_ia_j} = 0, \quad (150)$$

so

$$\boxed{\chi_{ij} = -H_y^{-1}(H_{a_ia_j} + H_{ya_j}\chi_i + H_{a_iy}\chi_j + H_{yy}[\chi_j, \chi_i])}. \quad (151)$$

The corresponding second total derivative of the pulled-back pencil is

$$D_{ij}T = \mathcal{T}_{a_ia_j} + \mathcal{T}_{ya_i}\chi_j + \mathcal{T}_{ya_j}\chi_i + \mathcal{T}_{yy}[\chi_i, \chi_j] + \mathcal{T}_y\chi_{ij}. \quad (152)$$

Thus even when $\mathcal{T}_{a_ia_j} = 0$ structurally, the re-equilibrated interaction vertex $D_{ij}T$ generally does not vanish. Equations (151)–(152) are the precise source of the “intrinsic” term in (111). Higher derivatives are obtained recursively by the multivariate implicit-function theorem; in implementation, automatic differentiation through the equilibrium solve or implicit Jacobian-vector products avoids forming high-order tensors explicitly.

APPENDIX B LOW-ORDER CONNECTED FORMULAS AND THE AFFINE CYCLE LIMIT

For reference, the first three complex connected derivatives are

$$D_i \log \det T = \text{tr}(GT_i), \quad (153)$$

$$D_{ij} \log \det T = \text{tr}(GT_{ij} - GT_iGT_j), \quad (154)$$

$$\begin{aligned} D_{ijk} \log \det T &= \text{tr}(GT_{ijk}) - \text{tr}(GT_iGT_{jk}) \\ &\quad - \text{tr}(GT_jGT_{ik}) - \text{tr}(GT_kGT_{ij}) \\ &\quad + \text{tr}(GT_iGT_jGT_k) + \text{tr}(GT_iGT_kGT_j). \end{aligned} \quad (155)$$

For the branch-free potential $\Psi = \log |\det T|$, take the real part of the right-hand sides on any nonsingular action chart.

A second proof of Corollary 5 makes the graph interpretation transparent. For the affine pencil $T = T_0(I + \sum_i a_i X_i)$,

$$\log \det T = \log \det T_0 + \sum_{p \geq 1} \frac{(-1)^{p+1}}{p} \text{tr} \left(\sum_i a_i X_i \right)^p. \quad (156)$$

The coefficient of the square-free monomial $\prod_{i \in S} a_i$ can appear only in the term $p = |S|$ and is

$$\frac{(-1)^{|S|+1}}{|S|} \sum_{\pi \in \mathfrak{S}_{|S|}} \text{tr}(X_{\pi_1} \cdots X_{\pi_{|S|}}). \quad (157)$$

Each cyclic orbit contains $|S|$ equal traces, producing (44). This proof also shows why repeated closed walks belong to the full logarithmic series while the distinct-action mixed derivative selects exactly the square-free connected return through the chosen action set.

APPENDIX C FINITE DIVIDENDS AS CUBICAL INTEGRALS

Let $\Delta_i^{\alpha_i}$ be the forward-difference operator

$$(\Delta_i^{\alpha_i} f)(a) = f(a + \alpha_i e_i) - f(a). \quad (158)$$

The operators commute for distinct coordinates. Repeated application of the fundamental theorem of calculus gives

$$\Delta_{i_1}^{\alpha_{i_1}} \cdots \Delta_{i_k}^{\alpha_{i_k}} f(0) = \int_{[0, \alpha_{i_1}] \times \cdots \times [0, \alpha_{i_k}]} D_S f(a_S) da_S. \quad (159)$$

Expanding the product of forward differences yields the Boolean-lattice Möbius sum in (46). Geometrically, the finite externality is therefore the integral of a connected interaction density over the action cube, not a value assigned to a single arbitrary intermediate operating point. For nonsmooth security objectives, the identity remains available piecewise on smooth cells, while the spectral-potential version can be used away from eigenvalue collisions.

APPENDIX D BLOCK CHORDLESS CYCLES AND COLLECTIVE OVERLAP CLOSURE

The MIMO quiver admits an exact isolated-cycle result.

Alternative derivation of Theorem 10.: Let Q_C have nonzero off-diagonal blocks only on one directed chordless cycle $C = (1, 2, \dots, m)$. Successive Schur elimination of unit diagonal blocks propagates the only nontrivial return around the cycle. After $m - 1$ eliminations, the remaining block is $I - (-1)^m \mathcal{H}_C$. Sylvester’s determinant identity handles unequal intermediate port dimensions, yielding exactly (69).

The isolated-cycle theorem must not be extrapolated to overlapping cycles. The smallest counterexample already appears with three scalar actions:

$$Q = \begin{bmatrix} 0 & a & c \\ b & 0 & 0 \\ d & 0 & 0 \end{bmatrix}, \quad \det(I + Q) = 1 - ab - cd. \quad (160)$$

There are two simple 2-cycles, $1 \leftrightarrow 2$ and $1 \leftrightarrow 3$, sharing action 1. Each isolated factor is nonsingular when $ab \neq 1$ and $cd \neq 1$, yet the combined core becomes singular when $ab + cd = 1$. For example, $ab = 0.6$ and $cd = 0.4$ produce collective closure while neither individual cycle closes. This is why the main text promotes *cycle clusters*/*SCC cores* rather than forcing every failure to have a unique single-cycle explanation.

APPENDIX E HOLONOMY INVARIANCE AND THE EXACT SCALARIZATION GATE

Under local basis changes $Q_{ij} \mapsto S_i^{-1} Q_{ij} S_j$, the holonomy of any closed cycle based at i transforms by conjugacy,

$\mathcal{H}_C \mapsto S_i^{-1} \mathcal{H}_C S_i$. Hence $\text{spec}(\mathcal{H}_C)$, $\text{tr}(\mathcal{H}_C^k)$, $\det \mathcal{H}_C$, and the characteristic polynomial are coordinate invariants. In contrast, a Frobenius norm of a commutator or a singular-value distance is invariant only under unitary-compatible port transformations; the main text therefore uses such quantities only after a physical port normalization is fixed.

Proposition 15 (Simultaneous scalarization). *A finite family of diagonalizable complex holonomies $\{\mathcal{H}_1, \dots, \mathcal{H}_q\}$ admits one common basis in which every member is diagonal if and only if the holonomies commute pairwise.*

Proof. Simultaneous diagonalization implies pairwise commutation immediately. Conversely, commuting diagonalizable matrices preserve one another's eigenspaces; diagonalize one member and recursively diagonalize the restrictions of the remaining commuting family on each eigenspace. $\square$

The physically stronger object is the generated algebra $\mathcal{A}_H = \text{alg}\{I, \mathcal{H}_1, \dots, \mathcal{H}_q\}$. Its invariant-subspace decomposition determines the smallest common block representation. An irreducible algebra obstructs exact common scalar channels, while a reducible algebra can still permit meaningful low-dimensional block strength descriptors.

APPENDIX F

A RIGOROUS DETERMINANT ERROR BOUND FOR CERTIFIED PRUNING

Let $Q = \tilde{Q} + E$ and assume $I + \tilde{Q}$ is nonsingular. Set

$$X = (I + \tilde{Q})^{-1} E, \quad \eta = \|X\|_2 < 1. \quad (161)$$

Then

$$\frac{\det(I + Q)}{\det(I + \tilde{Q})} = \det(I + X). \quad (162)$$

If r is the dimension of Q , every singular value of $I + X$ lies in $[1 - \eta, 1 + \eta]$. Therefore

$$\left| \log |\det(I + Q)| - \log |\det(I + \tilde{Q})| \right| \leq -r \log(1 - \eta). \quad (163)$$

This bound is deliberately conservative but branch free. It converts a transport-pruning tolerance into an explicit error bar on the pure-interaction spectral potential. A frequency sweep uses the supremum of the right-hand side over the protected band. If the resolvent becomes delocalized and η does not fall below the tolerance, the theory does not manufacture sparsity: the action core remains global.

APPENDIX G

GOHBERG–SIGAL INTERPRETATION OF THE EMT BOUNDARY CERTIFICATE

Theorem 14 is a finite-dimensional instance of the operator-valued Rouché theorem of Gohberg and Sigal [76]. Let $T_c(s)$ and $\Delta_c(s)$ be analytic (or finitely meromorphic with the standard zero-minus-pole bookkeeping) on and inside a contour Γ . If

$$\|T_c^{-1}(s)\Delta_c(s)\| < 1, \quad s \in \Gamma, \quad (164)$$

then T_c and $T_c + \Delta_c$ have the same logarithmic-residue count inside the contour. This theorem is stronger than comparing

individual eigenvalues after reduction: it certifies that the model boundary cannot create or remove a critical mode in the entire protected region. The contribution of the present paper is the identification of Δ_c with an *exterior self-energy error* generated by an EMT/RMS boundary choice.

APPENDIX H

WEIGHTED GEOMETRY OF DISTANCE TO FAILURE AND RECOVERY

Let a smooth scalar security margin $m(a)$ satisfy $m > 0$ in the safe region and $m = 0$ on a regular boundary. Equip action space with the engineering metric

$$g(a) = W(a)^T W(a) \succ 0. \quad (165)$$

The local weighted normal is $g^{-1} \nabla m$, and the first-order distance from a_0 to the boundary is

$$r_{\text{lin}}^-(a_0) = \frac{m(a_0)}{\|W^{-T} \nabla m(a_0)\|_2}. \quad (166)$$

The corresponding first-order most-dangerous action direction is proportional to $-g^{-1} \nabla m$. Curvature of the discriminant, obtained from the Hessian of m restricted to the tangent space, quantifies when this linear radius ceases to be reliable. The exact r^- and r^+ in (94)–(95) are therefore geodesic/optimization problems on the physically scaled action manifold, while (166) provides an interpretable initialization rather than a replacement for the finite solve.

APPENDIX I

LOW-RANK PORT CONTRACTIONS AND COMPUTATIONAL REUSE

Suppose $T_i = U_i C_i V_i^*$ at one operating point. Rather than form $G = T^{-1}$, solve

$$T X_i = U_i. \quad (167)$$

Then

$$\text{tr}(G T_i) = \text{tr}(C_i V_i^* X_i), \quad (168)$$

and a pair feedback term is

$$\text{tr}(G T_i G T_j) = \text{tr}(C_i V_i^* X_j C_j V_j^* X_i), \quad (169)$$

where the trace is taken only over the small port spaces. The same contractions generate MIMO quiver blocks. Across many frequencies, one can reuse sparse factorizations, rational Krylov subspaces, or contour-eigensolver factorizations. The resulting computational architecture is therefore compatible with large sparse descriptor models even though the mathematical derivation is written using G for clarity.

APPENDIX J

PROOF DETAILS FOR THE CONNECTED PARTITION-CYCLE FORMULA

For completeness, the first three mixed derivatives are written explicitly. Let $G = T^{-1}$ and use subscripts for total action derivatives on the equilibrium manifold:

$$D_i \log \det T = \text{tr}(GT_i), \quad (170)$$

$$D_{ij} \log \det T = \text{tr}(GT_{ij} - GT_i GT_j), \quad (171)$$

$$\begin{aligned} D_{ijk} \log \det T = & \text{tr}(GT_{ijk}) \\ & - \text{tr}(GT_{ij} GT_k + GT_{ik} GT_j + GT_{jk} GT_i) \\ & + \text{tr}(GT_i GT_j GT_k + GT_i GT_k GT_j). \end{aligned} \quad (172)$$

These are exactly the $|S| = 1, 2, 3$ cases of Theorem 4. Notice that nonlinear re-equilibration appears in T_{ij} and T_{ijk} ; in a frozen affine model they vanish and only closed resolvent products remain.

APPENDIX K

ARGUMENT-PRINCIPLE MODE COUNT WITHOUT A COMPLEX COOPERATIVE GAME

Let Γ be a contour on which $T(s, a)$ is nonsingular. The integer

$$N_\Gamma(a) = \frac{1}{2\pi j} \oint_\Gamma \partial_s \log \det T(s, a) ds \quad (173)$$

counts poles/zeros with the usual bookkeeping for the adopted nonlinear pencil. Differences $N_\Gamma(a_1) - N_\Gamma(a_0)$ are branch independent. If attribution of created/destroyed mode count is required, perform Möbius inversion on the integer-valued coalition function $N_\Gamma(S)$ itself rather than on a multivalued complex logarithm.

APPENDIX L

VARIATIONAL EXTENSION BEYOND ONE LINEARIZATION

The main claims are deliberately small-signal/re-equilibrated. Nevertheless, the same mechanism language has a natural nonlinear variational extension that clarifies what a future large-signal theory would require.

Let a nonlinear descriptor model, after regularization on a trajectory, induce the variational system

$$\dot{\xi} = A(t)\xi + B(t)u, \quad y = C(t)\xi, \quad (174)$$

with state-transition matrix

$$\Phi(t, \tau) = \mathcal{T} \exp\left(\int_\tau^t A(\sigma) d\sigma\right). \quad (175)$$

For local action/feedback ports one can define a finite-time transport

$$K_{ij}(t, \tau) = V_i^*(t) \Phi(t, \tau) U_j(\tau). \quad (176)$$

Products of such transports around a prescribed sequence form time-ordered finite-time return operators. They reduce to the frequency-domain quiver near a stationary LTI operating point, but in general they depend on the disturbance trajectory and traversal times.

This observation is important mainly as a scope boundary. A nonlinear fault-recovery/current-limit event cannot be certified

merely by evaluating the stationary holonomy $\mathcal{H}_C(j\omega)$. Large-signal claims require trajectory-dependent variational transport, hybrid-mode transitions, and possibly contraction/dissipativity arguments. The present EMT workflow therefore uses the small-signal connected core as a *scenario-selection and model-boundary hypothesis* and then validates the actual nonlinear trajectory in the detailed simulator. This avoids turning an elegant linear mechanism into an unjustified universal nonlinear stability theorem.

APPENDIX M

REPRODUCIBILITY REQUIREMENTS

A submission-ready implementation should archive every action definition, base and re-solved operating point, controller model/version, frequency grid, action path, numerical derivative step, pruning tolerance, SCC/cycle-cluster output, coalition metric, random seed, and full-vs-reduced validation trace. The reduced IEEE 39-bus mechanisms in this manuscript are not labeled manufacturer-grade. The intended falsification discipline is that every proposed shortcut—frozen action, pruned edge, reduced exterior, or local strength proxy—must be checked against the next more complete layer on the cases where the method claims a decision.

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